# ICOHP Descriptors and Thermodynamic Stability Across a 394-Structure Dataset

**Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis**

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*The project's central question is viability discrimination based on Δ(ICOHP) (§??), across a 394-structure dataset.*

## Contents

# 1 Objective and Problem Statement

**Central research question (see §??): does Δ(ICOHP) — the reaction-ICOHP descriptor for a compound's decomposition into elements, and specifically the sign-based endobondic/exobondic classification it supports — distinguish thermodynamically stable, metastable, and unstable compounds?**

# 2 Methodology

## 2.1 Dataset Selection

Materials Project query: unary and binary chemical systems, non-magnetic (|total magnetization| $\leq$ 0.01 $\mu_B$/formula unit — MP's categorical `magnetic_ordering` field is "Unknown" for ∼98% of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements. The dataset combines two complementary selection tracks: a first set of 186 compounds ($\leq$ 20 atoms/cell, 60 per stability family, a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system), and a second set of 163 compounds/elements ($\leq$ 40 atoms/cell) specifically targeting additional elemental references, off-equilibrium polymorphs (high-pressure phases, allotropes), and alkali/alkaline-earth binaries against N/O/F/P/S/Cl, chosen to build genuine same-composition polymorph groups and to test descriptors far from equilibrium; and a third set of 45 binary compounds, each pairing an experimentally known, maximally metastable compound (farthest above the convex hull among its chemical system) with 1–3 theoretical polymorphs of the same formula sitting even farther above the hull, a robustness test deliberately targeting stability edge cases.

## 2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by default by the `classify()` function (elemental composition + `is_metal`). A second, independent classification is built directly from each compound's ICOHP/ICOBI data (§??): `is_metal` for metallic character, then the mean ICOBI per bond (bond order, covalency) to separate ionic from covalent among non-metallic compounds. The two classifications are compared in §??.

## 2.3 VASP + LOBSTER Computation

Same template across all structures (static run, `ISYM=`−1, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`): `NBANDS` is derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

## 2.4 ICOHP/ICOBI Quantities: Integration, Cutoffs, Units

**LOBSTER generation (`lobsterin`, identical across the dataset)**: `pbeVaspFit2015` basis; COHP integration window from −35 to +5 eV around the Fermi level (`COHPstartEnergy`/`COHPendEnergy`); bond detection by interatomic distance from 0.1 to 6.0 Å (`cohpGenerator`, orbital-resolved); 0.05 eV Gaussian smearing (`gaussianSmearingWidth`).

**Integrated ICOHP/ICOBI (`ICOHPLIST.lobster`/ `ICOBILIST.lobster`).** A bond's integrated ICOHP is, by LOBSTER's own definition, the integral of COHP($E$) from the bottom of the window above up to the Fermi level (occupied states only), in eV; negative = bonding, positive = antibonding. ICOBI is the same integral over COBI (bond order index), dimensionless.

Per-compound aggregation (`percolation_path.py`): `icohp_sum`/`icobi_sum` over every symmetrically inequivalent bond label in `ICOHPLIST.lobster`/`ICOBILIST.lobster` (each periodic bond counted once, not once per direction); `icohp_mean`/ `icobi_mean`, the corresponding per-bond average — this intensive version, comparable across compounds of different cell sizes, is what is reported in Table **??** and the "ICOHP"/"ICOBI" columns of Appendix **??**. These quantities feed §**??** and the classic correlations of §**??**.

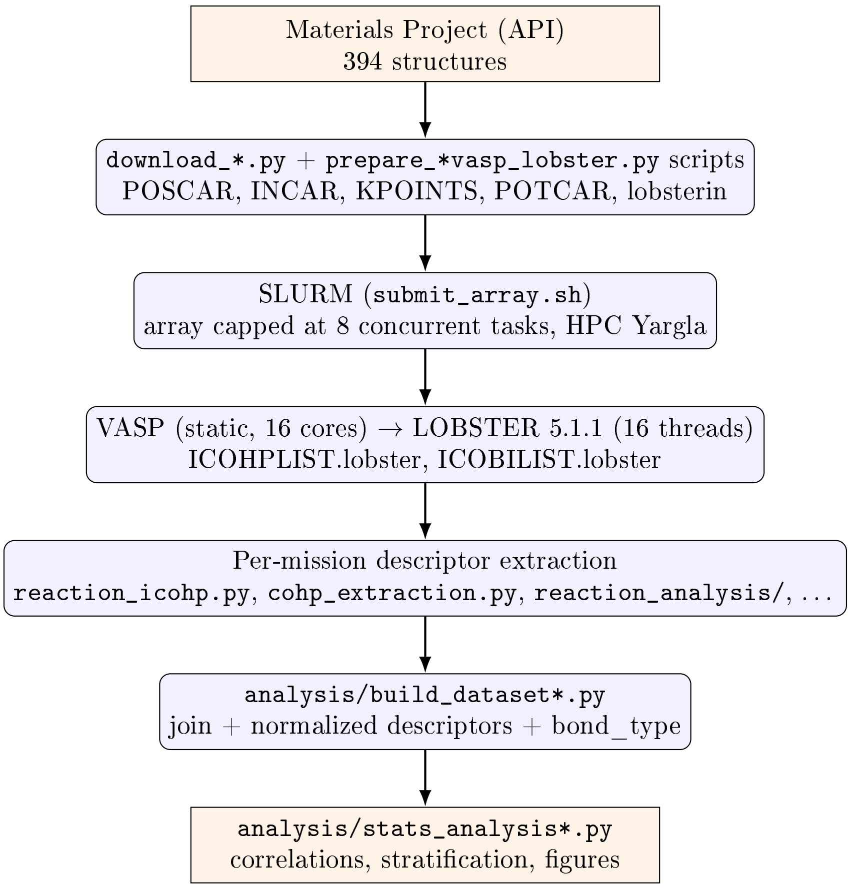
**Reaction Δ(ICOHP) (`reaction_analysis/`).** For a balanced reaction (typically compound → elements), $\Delta$ = sum of products' ICOHP − sum of reactants' ICOHP, "products − reactants" convention, in three normalizations: per formula unit (extensive), per transferred atom (the normalization used throughout this report unless noted), and per bond (diagnostic only, non-conservative — bonds are not conserved between differently-bonded structures). Positive sign ("endobondic"): breaking the reactant's bonds costs more than the products' bonds recover — a possible kinetic barrier. Negative sign ("exobondic"): no such barrier is visible from bonding alone.

**Antibonding population near the Fermi level (`cohp_extraction.py`) — a quantity distinct from the Δ(ICOHP) above.** Rather than integrating the whole occupied COHP, this quantity integrates only its antibonding part (COHP$>0$) over a one-sided window $(E_{\mathrm{ref}} - \Delta E,\ E_{\mathrm{ref}}]$, where $E_{\mathrm{ref}} = E_F$ (metallic compounds) or the VBM (gapped compounds), and $\Delta E \in \{0.5, 1.0, 2.0\}$ eV ($\Delta E = 1.0$ eV as the primary window) — only occupied states can destabilize the ground state via antibonding, hence the one-sided rather than symmetric window. Reported both raw (same "eV" unit convention as LOBSTER, though it is a Hamilton population integral, not literally an energy) and normalized by the total occupied ICOHP magnitude at the same $E_{\mathrm{ref}}$. This quantity, and only this one, feeds the descriptor in §**??** — the reaction Δ(ICOHP) (§**??**–**??**) integrates the whole occupied COHP with no restriction to a window near $E_F$.

## 2.5 Statistics

Spearman correlation (no reason to expect a linear relationship) between each ICOHP/ICOBI descriptor and a stability target (`energy_above_hull` or `formation_energy_per_atom` depending on the descriptor), overall and per subgroup (`bond_type`, `is_metal`); any $n < 15$ subgroup is explicitly flagged, never presented bare. **No SISSO, no symbolic regression** at this stage (see §**??**).

## 3 Workflow Diagram




## 4 Environment

Identical across all 394 structures (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`, `pydantic`, `pyyaml` (see `requirements.txt`). Per-step core allocation: §**??**.

## 5 Dataset

The full set of computed compounds and elements totals **394 structures**.

Table 1: Dataset composition by element count.

| Composition | $n$ |
|---|---|
| Single-element references | 68 |
| Multi-element compounds | 326 |
| **Total** | **394** |

**Bond type.** Two independent classifications (§2.2): the `classify()` function (elemental composition + `is_metal`) and a classification built directly from each compound's ICOHP/ICOBI data (`is_metal` for metallic character; otherwise the ICOBI of the **dominant first-coordination-shell** species pair — not a mean over every neighbor LOBSTER reports within the wider cutoff used for the percolation graph, which strongly dilutes the signal for dense covalent solids such as diamond — compared against a threshold calibrated at the midpoint of the median ICOBI of

Table 2: Stability class (by $E_{\mathrm{hull}}$ and experimental/theoretical character).

| Class | $n$ | $E_{\mathrm{hull}}$ range (eV/atom) |
|---|---|---|
| stable ($E_{\mathrm{hull}} = 0$) | 124 | 0.0000 |
| experimental metastable | 137 | 0.0000347 – 2.1080 |
| theoretical metastable | 131 | 0.0000843 – 3.1688 |
| unknown (COD reference, no $E_{\mathrm{hull}}$) | 2 | — |
| **Total** | **394** | |

compounds `classify()` already labels ionic/covalent, i.e. 0.5052, to separate ionic from covalent among non-metallic compounds). A fourth label, **mixed** (Zintl-type), identifies compounds where a *homoatomic* pair above the covalent threshold (e.g. N–N in an azide, P–P in a polyphosphide, S–S in a polysulfide) coexists with a markedly weaker *heteroatomic* pair linking a distinct cation — the structural signature of Zintl–Klemm chemistry, where a single ionic/covalent label would be misleading.

Table 3: Bond type: `classify()` vs. the ICOHP/ICOBI classification (concordance on the subset `classify()` labels: 161/181 = 89.0%).

| `classify()` \ ICOHP/ICOBI | metallic | ionic | covalent | mixed | unclassified |
|---|---|---|---|---|---|
| metallic | 88 | 0 | 0 | 0 | 0 |
| ionic | 0 | 45 | 5 | 3 | 0 |
| covalent | 0 | 10 | 28 | 2 | 0 |
| unclassified | 138 | 31 | 19 | 23 | 0 |
| **Total** | **226** | **86** | **52** | **28** | **0** |

Of the 211 compounds `classify()` leaves "unclassified" (essentially alkali/alkaline-earth $\times$ {N, P, S} pairs outside `IONIC_ANIONS`, and metals bonded to an anion-like element that `classify()` deliberately excludes from its "metallic" bucket), the ICOHP/ICOBI classification resolves **50** into ionic (31) or covalent (19), and identifies **23** as mixed (Zintl-type); the remaining 138 are metallic, already covered by `is_metal`. Where both methods render a verdict (metallic/ionic/covalent), agreement is strong (89.0%) but not total: 15 ionic/covalent disagreements, and 5 compounds `classify()` labels ionic or covalent but that the ICOBI classification identifies as mixed (their dominant homoatomic bond coexists with a markedly weaker heteroatomic link) — worth checking case by case before using either classification as a stratification variable.

# 6 Results

## 6.1 Correlations (Spearman) — classic ICOHP aggregates

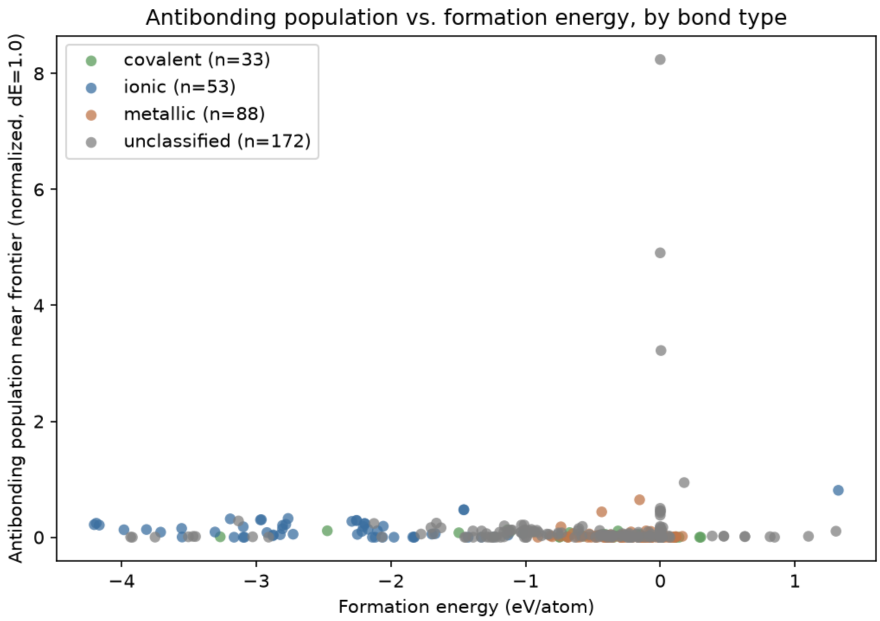
Table 4: Spearman correlations, classic ICOHP aggregates (sum, minimum) against $E_{\mathrm{hull}}$, 186-compound subset.

| Group | Metric | $n$ | $\rho$ | $p$ |
|---|---|---|---|---|
| all | ICOHP sum | 186 | 0.032 | 0.670 |
| all | ICOHP min | 186 | 0.093 | 0.209 |
| metallic | ICOHP sum | 88 | 0.223 | 0.037 |
| metallic | ICOHP min | 88 | 0.198 | 0.065 |

These four rows serve as a reference baseline: raw ICOHP aggregates (per-compound sum, minimum) show little to no correlation with $E_{\mathrm{hull}}$, except within the metallic subgroup. The descriptors built further on (antibonding population, reaction ICOHP, §??–§??) do markedly better, against `formation_energy_per_atom` rather than `energy_above_hull`.

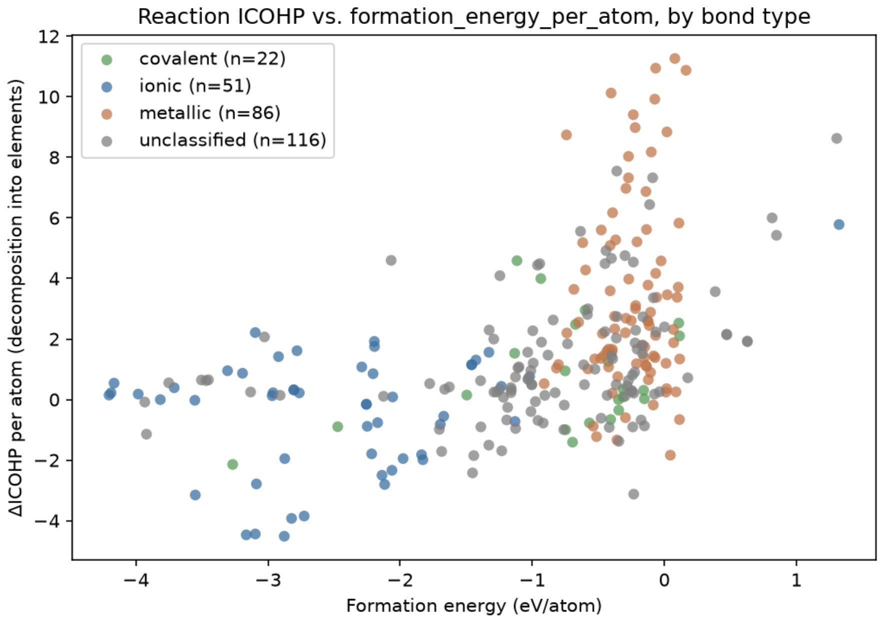
## 6.2 Antibonding Population Near the Fermi Level

A question distinct from the fully-integrated ICOHP (§2.4): not *how much* bonding a compound has, but *how* its COHP is distributed in energy — the occupied antibonding fraction just below the Fermi level/VBM, by analogy with Peierls/Jahn–Teller-type electronic instabilities. Result, against `formation_energy_per_atom`: $\rho = -0.282$, $p = 9.2 \times 10^{-8}$ ($n = 346$). The sign is the *opposite* of the naive Peierls/Jahn–Teller reading. The `bond_type=covalent` subgroup ($n = 33$) survives stratification clearly ($\rho = -0.551$, $p = 0.0009$) — the project's most robust stratified result; `is_metal=False` also survives ($n = 157$, $p = 0.001$). An earlier result on the `bond_type=ionic` subgroup ($n = 9$, $\rho = -0.683$) does not replicate once the sample grows to $n = 53$ ($\rho = -0.184$, not significant) — a clear demonstration of the danger of $n < 15$ subgroups.




## 6.3 Reaction ICOHP

An ICOHP analog of `formation_energy_per_atom`: is a compound's total ICOHP "cheaper" than that of the same atoms as reference elements? Result: $\rho = 0.502$, $p = 6.3 \times 10^{-19}$ ($n = 275$) — the dataset's strongest and most robust continuous correlation, with an intuitive sign (less bonding than in elemental form ⇒ less stable formation). `bond_type` stratification fully explains this layer of the pooled signal (Kruskal–Wallis $p = 1.1 \times 10^{-15}$, no stratum survives) — **but `is_metal` stratification survives on both sides** (`is_metal=False`: $\rho = 0.319$, $p = 0.0001$; `is_metal=True`: $\rho = 0.256$, $p = 0.0025$): between-group clustering fully explains the `bond_type`-level grouping, but not a real relationship within a fixed `is_metal`. Case 2 (direct same-formula polymorph comparison, 49 groups): 53.1% agreement between the most-bonded and the most-stable member, against a 45.9% chance baseline (exact Poisson-binomial test, $P = 0.19$) — no signal.




**Still no SISSO.** Reaction ICOHP is the dataset's strongest raw correlate of `formation_energy_per_atom`, and the only descriptor whose signal is not fully explained away by between-group clustering at every level tested — the best argument yet for a stratified (`is_metal`-conditioned) rather than pooled feature search, still not attempted. Detailed reports: `analysis/REPORT_antibonding.md`, `analysis/REPORT_reaction_icohp.md`.

## 6.4 Δ(ICOHP)-Based Viability Discrimination

Full detail: `analysis/REPORT_delta_icohp_viability.md`, `analysis/test_delta_icohp_viability.py`.

**Reference validation.** Per-bond-type ICOHP values for 7 reactions with independently known ΔICOHP (transcribed independently of any calculation in this project) are run end to end through `balance.py`/`delta.py`/`classify.py`:

Table 5: Validating the endobondic/exobondic classification on 7 reactions with independently known ΔICOHP.

| Reaction | Reference (kJ/mol) | Computed (kJ/mol) | Label Ref. | Label Computed |
|---|---|---|---|---|
| $Pb(N_3)_2 \rightarrow Pb + 3\,N_2$ | +1345 | +1344.3 | endo | endo |
| $S_4N_2 \rightarrow \frac{1}{2}S_8 + N_2$ | +258 | +258.5 | endo | endo |
| $S_4N_4 \rightarrow \frac{1}{2}S_8 + 2N_2$ | +399 | +397.3 | endo | endo |
| ZnSn → Zn + Sn | −337 | −336.6 | exo | exo |
| CaO[sphalerite] → CaO[rocksalt] | −79 | −78.9 | exo | exo |
| $CaN \rightarrow \frac{1}{3}Ca_3N_2 + \frac{1}{6}N_2$ | −205 | −204.8 | exo | exo |
| $Mn_2O_7 \rightarrow 2MnO_2 + \frac{3}{2}O_2$ | −186 | −187.7 | exo | exo |

The 7 cases reproduce the reference ΔICOHP to within a few kJ/mol (source per-bond value rounding) and every `BondingLabel` matches. The $Mn_2O_7$ case is a useful guardrail on its own: exobondic under this static sign criterion, yet $Mn_2O_7$ is a real, observed compound that simply decomposes slowly (gradual $O_2$ loss) rather than through abrupt bond breaking — `classify.py` structurally attaches a warning to *every* `UNSTABLE_NONEXISTENT` verdict for exactly this reason, rather than special-casing $Mn_2O_7$.

This validation only compares the arithmetic of the manuscript's own published ICOHP values, independent of any local crystal structure — Appendix ?? documents, for the 16 species involved, which MP/COD structure was used in this dataset and whether it could be verified to match the manuscript's description (3 real corrections identified: $N_2$, the $Pb(N_3)_2$ polymorph, ZnSn's competing tin phase). For $N_2$ and ZnSn, a DFT+LOBSTER calculation of this project's own (not just the manuscript's arithmetic) was additionally compared directly: $N_2$ reproduces the manuscript's ICOHP to within 0.7%; ZnSn reproduces the same sign (bonding-favorable decomposition) but with ~1.9× larger magnitude — a gap judged not significant at this stage, since a bulk intermetallic's ICOHP sum is far more sensitive to basis/pseudopotential/cutoff choices than a simple gas dimer.

Correlation alone (Pearson/Spearman) is not the right test for "do our numbers match the reference values" — it is satisfied by any monotonic relationship, even with a large systematic offset or scale factor. **Lin's concordance correlation coefficient** (CCC) specifically tests agreement with the identity line (computed = reference), the claim actually being tested here:

Table 6: Computed-vs-reference concordance on the 7 validation reactions.

| Statistic | Value |
|---|---|
| $n$ | 7 |
| Lin's CCC | **0.999998** |
| Pearson $r$ | 0.999999 ($p = 3.5 \times 10^{-15}$) |
| Sign agreement | 7/7 |
| Mean \|residual\| | 0.76 kJ/mol |
| Max \|residual\| | 1.70 kJ/mol |
| RMSE | 0.98 kJ/mol |

A CCC this close to 1 means the two methods agree essentially to the reporting precision of the reference values themselves (rounded to 2–3 decimals), not merely that they move together — the implementation is validated, not just spot-checked.

**Applied to the full dataset, pooled. 281 case-1 reactions**, every element and compound with a computed decomposition-into-elements reaction, pooled into a single population (grouping by selection provenance would be exactly the kind of between-group confound this project's methodology is meant to catch).

Table 7: Spearman correlations, ΔICOHP/atom (case 1, 281 reactions) against two continuous stability targets.

| Target | $n$ | $\rho$ | $p$ |
|---|---|---|---|
| `formation_energy_per_atom` | 275 | −0.5016 | $6.3 \times 10^{-19}$ |
| `energy_above_hull_eV_per_atom` | 279 | −0.1096 | 0.068 |

(Sign: `reaction_analysis`'s products−reactants convention, opposite to `reaction_icohp.py`'s own column §?? — same magnitude, flipped sign.) The formation-energy result is the same relationship as §??'s, confirmed here on the pooled population.

**Sign-based discrimination (endobondic/exobondic).** Experimental viability indicator (`theoretical=False` means experimentally realized):

Fisher's exact test: $p = 0.10$, odds ratio = 1.72 (endobondic is ~ 1.7× more frequent among experimentally realized compounds — the expected direction, endobondic being designed as a kinetic-barrier signature — but not significant at this sample size).

**Reading trap, explicitly flagged by `classify.py` itself**: Table ?? shows 130 *experimental* (i.e. genuinely existing) compounds labeled *exobondic*. This is **not** a contradiction —

Table 8: Endobondic/exobondic label against experimental/theoretical status.

| | endobondic | exobondic |
|---|---|---|
| experimental ($n = 177$) | 47 | 130 |
| theoretical only ($n = 98$) | 17 | 81 |

exobondic only implies "not viable" once combined with `delta_energy` (`classify_viability()`, §??): a compound is only `UNSTABLE_NONEXISTENT` if decomposition into elements is *both* exobondic *and* thermodynamically favorable ($\Delta E < 0$), which is rare (decomposing a real compound into pure elements is almost never favorable). Verified directly on the same population as Table **??**:

Table 9: Computed `ViabilityLabel` (`classify_viability()`) on the same population as Table **??** ($n = 281$).

| | experimental ($n = 177$) | theoretical ($n = 98$) |
|---|---|---|
| `STABLE_ON_HULL` | 170 | 76 |
| `UNSTABLE_NONEXISTENT` | 7 | 20 |
| `METASTABLE_VIABLE` | 0 | 2 |

6 additional reactions (COD-sourced, no `formation_energy_per_atom`) excluded, insufficient data.

**Of the 130 experimental exobondic compounds, only 7 are predicted `UNSTABLE_NONEXISTENT`** — the other 123 are `STABLE_ON_HULL`, perfectly consistent with their real existence despite the exobondic label. The sign alone (endobondic/exobondic, Table **??**) only becomes a viability prediction once combined with thermodynamics — reading Table **??** in isolation as "exobondic = not viable" gives a misleading picture.

Table 10: Median ΔICOHP/atom and % endobondic by stability class ($n = 175$).

| Class | $n$ | median ΔICOHP/atom (eV) | % endobondic |
|---|---|---|---|
| experimental metastable | 59 | $-1.077$ | **27.1%** |
| stable | 59 | $-1.589$ | 20.3% |
| theoretical metastable | 59 | $-1.659$ | **11.9%** |

Kruskal–Wallis across the three groups: $H = 2.81$, $p = 0.245$ — **not significant**, but the ordering is exactly what physics predicts: experimentally metastable compounds (known to exist despite sitting off the convex hull — exactly the category the endobondic/exobondic framework addresses) have the highest endobondic fraction; never-synthesized theoretical metastable compounds have the lowest.

**Reading.** The 7 reference reactions reproduce almost exactly — the implementation is correct. Extending the same sign-based logic to a much larger, more heterogeneous, and generally much more comfortably stable population does not yet produce a statistically significant stable/metastable/unstable discriminant, even though every trend points the right way. This is not a contradiction: the 7 reference reactions were specifically chosen as stability edge cases ($Pb(N_3)_2$, $S_4N_2/S_4N_4$, ZnSn, $Mn_2O_7$) where a kinetic-barrier explanation is specifically needed; most of this project's 281 compounds are not — they are comfortably stable or comfortably not, where the thermodynamic driving force (the formation-energy correlation above) can simply dominate the sign-based kinetic signal. The full `ViabilityLabel` machinery in `classify.py` (beyond the sign-based `BondingLabel` alone used here) needs exactly such a near-stability-boundary population — for the current dataset, decomposition into elements is strongly exothermic for almost every compound (normal for stable inorganic chemistry), which would trivially classify nearly everything `STABLE_ON_HULL` and add no information beyond what `energy_above_hull` already says directly.

## 7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the first 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source (the fixes, applied to the shared SLURM/INCAR templates, carry over to the whole dataset):

- **$Al_{12}W$, BW, $WO_2$, TcW, TiW** (all tungsten-containing): the local mirror's `W_sv` POTCAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).
- **$Al_{12}W$** failed again after that fix, this time with a VASP SIGSEGV — a known issue (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`'s per-compound SLURM template. Fixed; recovered on retry.

Final success rate across all 394 structures computed to date: **394/394**.

## 8 Compute Cost

**394 structures computed in total**: 68 single-element references and 326 multi-element compounds (§??).

Every VASP job, then LOBSTER job, runs on the **Yargla** HPC cluster, within a single SLURM allocation of one node (`-nodes=1`) and **16 cores**: VASP under MPI parallelism (`-ntasks=16 -cpus-per-task=1`, `srun`), then LOBSTER under OpenMP parallelism on the same 16 cores (`OMP_NUM_THREADS=16`, single process, run sequentially after VASP within the same job). The SLURM array (`submit_array.sh`) is capped at 8 concurrent tasks.

Table 11: VASP and LOBSTER compute time (n=349; timing not available for every compound).

| | VASP | LOBSTER |
|---|---|---|
| $n$ | 349 | 348* |
| Mean (s) | 168 | 404 |
| Median (s) | 58 | 110 |
| Max (s) | 5715 (1.6 h) | 12063 (3.4 h) |
| Sequential total | 16.3 h | 39.0 h |

* 1 of 349 structures had no usable LOBSTER step (trivial-cell reference element).

Combined sequential total (VASP + LOBSTER, 349 structures): **55.3 h**. Given the 8-concurrent-task cap, the order of magnitude for actual elapsed time to process this volume as a single batch would be ∼6.9 h — a rough estimate (dividing by 8, ignoring queue time) given for context.

## 9 Current Limits and Next Steps

- **`classify()` coverage**: 211/394 structures (54%) fall outside the ionic/covalent/metallic categories, mostly alkali/alkaline-earth × {N, P, S} bonds that `IONIC_ANIONS` does not recognize (§??) — purely informative here, but a real limitation for any mission that would want to stratify on it.

- **No true dynamical "unstable" class**: `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.
- **Per-subgroup sample size**: remains this project's structural limit at every new stratification (`bond_type`, `is_metal`) — see §?? for a concrete example where an $n < 15$ subgroup did not replicate once the sample grew.
- **Why not SISSO yet**: reaction ICOHP (§??) is now the strongest correlate and the first whose signal is not fully explained away by between-group clustering at every level tested — the best argument yet for a stratified feature search, still not attempted.

## 10 Conclusion

Reitz & Dronskowski's endobondic/exobondic methodology (§??) reproduces faithfully across the 7 independently known reference reactions (Lin's CCC 0.999998) and applies to this report's full dataset (281 decomposition-into-elements reactions). On that population, sign-based ΔICOHP discrimination points the expected way (endobondic more frequent among experimental compounds, and among metastable rather than stable or never-synthesized compounds) but does not reach statistical significance at this scale (§??). Reaction ICOHP otherwise remains the dataset's strongest and most robust continuous correlate against formation energy ($\rho = 0.502$, §??).

Two concrete directions follow. First, deliberately extending to compounds near the stability boundary, where a kinetic-barrier explanation is specifically needed, rather than letting the dataset grow incidentally from whatever other analyses require — most current compounds are comfortably stable or comfortably unstable, a regime where the thermodynamic driving force simply dominates the sign-based kinetic signal. Second, restricting the reaction Δ(ICOHP) to a window near the Fermi level rather than the fully integrated ICOHP, modeled on the antibonding population of §??: two variants of this construction are detailed in Appendix ?? (antibonding part only, and the full signed ICOHP restricted to the window) — the first survives every stratification tested, a result no other descriptor in this report reaches.

*Source code:* `https://github.com/John-Akapulco/viability`
*Data:* `analysis/percolation_vs_formation_energy.csv`
*Detailed reports:* `analysis/REPORT_*.md`

# Appendix (SI)

## A Compound and Element List

Complete list of the 392 compounds and elements with usable ICOBI data (of 394 total; 2 COD-sourced compounds lack `is_metal`/ICOBI, §??), organized by section according to the independent ICOHP/ICOBI classification of §?? (`is_metal` for metallic character; otherwise the ICOBI of the **dominant first-coordination-shell** species pair — not a mean over every neighbor LOBSTER reports within the wider percolation-graph cutoff — compared against a threshold of 0.5052 to separate ionic from covalent; a **mixed** (Zintl-type) category groups compounds where a strongly covalent homoatomic pair coexists with a markedly weaker heteroatomic pair, e.g. the azide ion $N_3^-$ ionically bound to its cation) — no "bond type" column; each section instead states its own observed ICOBI range. ICOBI and ICOHP are per-bond means (§2.4); the ICOHP/ICOBI antibonding populations are the normalized values at $\Delta E = 1.0$ eV (§??, §??).

No percolation-weight column here — kept only, for the original campaign's 186 compounds, in the duplicated table of Section ??.

## A.1 Metallic

Table S1: Metallic compounds and elements (`is_metal=True`, observed ICOBI 0.0000 to 2.9506) — formula, mp-id, space group, dominant (first-shell) bond ICOBI, dominant bond ICOHP, normalized ICOHP and ICOBI antibonding populations ($\Delta E$ = 1.0 eV).

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| Ag* | mp-124 | Fm-3m | 0.1545 | -1.2496 | 0.0000 | 0.0000 |
| AgN* | mp-1091417 | Cmce | 0.4624 | -3.4181 | 0.0088 | 0.0030 |
| AgN | mp-1009766 | F-43m | 0.4854 | -3.5762 | 0.0277 | 0.0173 |
| AgN | mp-1424734 | Ama2 | 0.8238 | -5.9446 | 0.0107 | 0.0025 |
| AgN | mp-1428312 | Pm-3m | 0.2177 | -1.4991 | 0.0083 | 0.0041 |
| Al* | mp-134 | Fm-3m | 0.2270 | -1.4193 | 0.0000 | 0.0000 |
| $Al_{12}W$* | mp-11227 | Im-3 | 0.5027 | -2.7492 | 0.0000 | 0.0000 |
| $Al_3Os_2$* | mp-16521 | I4/mmm | 0.3022 | -1.6476 | 0.0037 | 0.0000 |
| $Al_4Si$ | mp-1228120 | R-3m | 0.2614 | -2.0375 | 0.0087 | 0.0030 |
| AlNi | – | – | 0.2676 | -1.9848 | 0.0016 | 0.0023 |
| AlSb | mp-1023918 | Cmcm | 0.2647 | -1.2802 | 0.0000 | 0.0000 |
| As* | mp-158 | Cmce | 0.8848 | -3.9683 | 0.0103 | 0.0000 |
| $AsPd_2$* | mp-1080831 | P-62m | 0.3410 | -2.4185 | 0.0077 | 0.0157 |
| AsRh* | mp-22079 | Pnma | 0.3323 | -1.8639 | 0.0449 | 0.0158 |
| $AsRh_2$* | mp-1102364 | Pnma | 0.2438 | -1.3778 | 0.1158 | 0.0189 |
| Au* | mp-81 | Fm-3m | 0.1962 | -1.3381 | 0.0000 | 0.0000 |
| $B_3Ir_2$ | mp-1228647 | C2/m | 0.4473 | -3.3600 | 0.0047 | 0.0011 |
| $BPd_5$ | mp-1228665 | I4/mmm | 0.4754 | -3.9924 | 0.0973 | 0.0181 |
| BW* | mp-7832 | I4_1/amd | 0.4277 | -3.9556 | 0.0000 | 0.0000 |
| Ba* | mp-122 | Im-3m | 0.0002 | -0.0122 | 8.2400 | 2.9573 |
| $BaAu_2$* | mp-30363 | P6/mmm | 0.6152 | -3.0712 | 0.0144 | 0.0034 |
| $BaF_3$ | mp-1183303 | I4/mmm | 0.0350 | -0.5277 | 0.2805 | 0.2537 |
| $BaN_2$* | mp-1102371 | Cc | 0.1605 | -0.4471 | 0.1039 | 0.0214 |
| $BaSn_5$* | mp-571353 | P6/mmm | 0.2221 | -0.8507 | 0.0456 | 0.0042 |
| Be* | mp-87 | P6_3/mmc | 0.2033 | -1.8975 | 0.0000 | 0.0000 |
| $Be_{12}Pd$* | mp-12666 | I4/mmm | 0.2078 | -1.9466 | 0.0000 | 0.0000 |
| $Be_2Cu$* | mp-2031 | Fd-3m | 0.3616 | -3.0863 | 0.0000 | 0.0001 |
| $Be_2Nb_3$* | mp-11272 | P4/mbm | 0.4200 | -3.5436 | 0.0000 | 0.0000 |
| $Be_2V$* | mp-11281 | P6_3/mmc | 0.3658 | -2.7859 | 0.0000 | 0.0000 |
| $BeAu_3$ | mp-983539 | Pm-3m | 0.2230 | -1.9129 | 0.0000 | 0.0030 |
| BeCu | – | – | 0.1984 | -1.8359 | 0.0000 | 0.0000 |
| $BeF_2$ | mp-975644 | P6/mmm | 0.1548 | -1.6978 | 0.0863 | 0.0000 |
| $Bi_2O_3$* | mp-558953 | P2_1/c | 0.9517 | -6.7444 | 0.0328 | 0.0038 |
| $Bi_2O_3$ | mp-1383505 | P1 | 0.9506 | -7.1985 | 0.0609 | 0.0000 |
| C* | mp-48 | P6_3/mmc | 1.2195 | -11.5387 | 0.0000 | 0.0000 |
| Ca* | mp-21 | Im-3m | 0.0025 | -0.0304 | 3.2206 | 3.3021 |
| $Ca_3N_2$ | mp-1013524 | Pm-3m | 0.1107 | -0.3623 | 0.0090 | 0.0000 |
| $CaAl_4$* | mp-1749 | I4/mmm | 0.4595 | -1.8701 | 0.0086 | 0.0002 |
| $CaGa_2$* | mp-992 | P6/mmm | 0.9791 | -3.8887 | 0.0219 | 0.0000 |
| $CaGa_4$* | mp-2461 | C2/m | 0.4641 | -2.1536 | 0.0084 | 0.0001 |
| $CaGe_2$ | mp-643542 | C2/m | 0.7379 | -3.1645 | 0.0337 | 0.0038 |
| CaN* | mp-1058549 | Fm-3m | 0.1201 | -0.6934 | 0.0450 | 0.0656 |
| CaO | mp-1181903 | Fd-3m | 0.0725 | -0.5516 | 0.1865 | 0.0756 |
| CaSn* | mp-2450 | Cmcm | 0.7044 | -2.1965 | 0.0629 | 0.0276 |

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| Cd* | mp-94 | P6_3/mmc | 0.2423 | -1.6119 | 0.0000 | 0.0000 |
| CdPd* | mp-1696 | P4/mmm | 0.2022 | -1.5904 | 0.0000 | 0.0004 |
| Co* | mp-102 | Fm-3m | 0.2680 | -1.9150 | 0.0189 | 0.0373 |
| CoB* | mp-20857 | Pnma | 0.4568 | -4.5928 | 0.0099 | 0.0000 |
| $CoSe_2$ | mp-1390196 | Fd-3m | 0.7048 | -4.3267 | 0.0348 | 0.0079 |
| Cr* | mp-90 | Im-3m | 0.2487 | -0.6245 | 0.0115 | 0.0049 |
| $Cr_2C$ | mp-1226378 | P-3m1 | 0.5270 | -3.0803 | 0.0084 | 0.0000 |
| CrMo | mp-1226200 | Cmmm | 0.3600 | -1.3679 | 0.0081 | 0.0026 |
| Cs* | mp-1 | Im-3m | 0.0556 | -0.1230 | 0.1696 | 0.0693 |
| $CsO_2$* | mp-1441 | I4/mmm | 1.0942 | -11.4627 | 0.1143 | 0.1436 |
| $CsO_2$ | mp-1096936 | Cmmm | 1.1298 | -11.0556 | 0.1197 | 0.0872 |
| Cu* | mp-30 | Fm-3m | 0.0456 | -0.2488 | 0.1641 | 0.0946 |
| $Cu_3Ge$* | mp-19724 | Pmmn | 0.1369 | -0.8340 | 0.0963 | 0.0410 |
| Fe* | mp-13 | Im-3m | 0.1417 | -0.5370 | 0.0922 | 0.0794 |
| FeS* | mp-505531 | P4/nmm | 0.4967 | -2.5883 | 0.0736 | 0.0024 |
| $FeTe_2$* | mp-1102002 | Pa-3 | 0.4279 | -1.7861 | 0.0401 | 0.0000 |
| Ga* | mp-142 | Cmce | 0.4308 | -2.6378 | 0.0000 | 0.0000 |
| Ge* | mp-32 | Fd-3m | 0.8956 | -4.6450 | 0.0015 | 0.0000 |
| $Ge_2Os$* | mp-10032 | C2/m | 0.4042 | -2.1066 | 0.0069 | 0.0003 |
| $Ge_4Rh$* | mp-1104286 | P3_121 | 0.3355 | -1.8036 | 0.0034 | 0.0006 |
| Hf* | mp-103 | P6_3/mmc | 0.3121 | -1.1612 | 0.0000 | 0.0000 |
| HfMo | mp-1224314 | Cmmm | 0.3368 | -1.2722 | 0.0000 | 0.0000 |
| $HfTc_2$* | mp-1095669 | P6_3/mmc | 0.3775 | -1.4429 | 0.0098 | 0.0012 |
| Hg* | mp-10861 | P6/mmm | 0.1622 | -0.9652 | 0.0000 | 0.0000 |
| $Hg_4Pt$* | mp-936 | Im-3m | 0.5477 | -3.5588 | 0.0000 | 0.0005 |
| $IN_2$ | mp-1097011 | I4/mcm | 2.9506 | -23.1289 | 0.0177 | 0.0275 |
| In* | mp-1055994 | I4/mmm | 0.2177 | -1.1562 | 0.0000 | 0.0000 |
| $In_3Ir$* | mp-630976 | P4_2/mnm | 0.5910 | -3.4701 | 0.0000 | 0.0000 |
| $InSe_2$ | mp-1232036 | P4/nmm | 0.2771 | -1.4780 | 0.0349 | 0.0038 |
| Ir* | mp-101 | Fm-3m | 0.3469 | -2.4130 | 0.0317 | 0.0274 |
| K* | mp-10157 | Fm-3m | 0.0430 | -0.0808 | 0.0923 | 0.0499 |
| $K_2O$ | mp-1223784 | P4/mmm | 0.1375 | -0.3763 | 0.0831 | 0.0000 |
| $K_5As_4$ | mp-684799 | P-1 | 0.9859 | -3.9306 | 0.0452 | 0.0189 |
| $KN_2$* | mp-1071868 | Cmc2_1 | 1.0490 | -7.9889 | 0.0202 | 0.0000 |
| $KN_2$ | mp-1180757 | Cmcm | 1.0210 | -7.4152 | 0.0411 | 0.0000 |
| $KN_3$ | mp-636056 | I4/mcm | 2.8711 | -22.8747 | 0.0090 | 0.0107 |
| KP* | mp-1058315 | R-3m | 0.1254 | -0.4533 | 0.0190 | 0.0442 |
| KP | mp-1057787 | Immm | 0.1070 | -0.4621 | 0.0170 | 0.0218 |
| $KSn_3$ | mp-1185151 | P6_3/mmc | 0.4006 | -1.5431 | 0.0179 | 0.0027 |
| Li* | mp-1018134 | R-3m | 0.0886 | -0.6931 | 0.0000 | 0.0000 |
| $Li_{13}Sn_5$* | mp-30769 | P-3m1 | 0.6069 | -2.7881 | 0.0000 | 0.0000 |
| $Li_3Ag$ | mp-976408 | I4/mmm | 0.1084 | -1.0444 | 0.0000 | 0.0000 |
| $Li_3Hg$* | mp-1646 | Fm-3m | 0.1488 | -1.2194 | 0.0000 | 0.0000 |
| LiB* | mp-1001835 | P6_3/mmc | 1.1571 | -9.9815 | 0.0000 | 0.0000 |
| $LiTl_3$ | mp-1185253 | I4/mmm | 0.2370 | -1.2989 | 0.0000 | 0.0000 |
| Mg* | mp-153 | P6_3/mmc | 0.0000 | 0.0000 | – | – |
| $Mg_{15}B$ | mp-1023515 | P-6m2 | 0.0280 | -0.1152 | 0.9423 | 0.5909 |
| $Mg_2Ni$* | mp-2137 | P6_222 | 0.0818 | -0.2478 | 0.6457 | 0.3329 |
| $MgPd_3$* | mp-7753 | Pm-3m | 0.2542 | -1.7837 | 0.0264 | 0.0304 |
| MgSn | mp-1094669 | Pm-3m | 0.2238 | -0.7050 | 0.1000 | 0.0334 |
| MgSn | mp-1094801 | P4/mmm | 0.3134 | -1.2760 | 0.1099 | 0.0281 |
| Mn* | mp-35 | I-43m | – | -0.6082 | 0.0411 | 0.0047 |
| $MnAl_{12}$* | mp-1104165 | Im-3 | 0.2798 | -1.6283 | 0.0000 | 0.0000 |
| $MnO_2$* | mp-510408 | P4_2/mnm | 0.4433 | -3.0527 | 0.0548 | 0.0603 |
| Mo* | mp-129 | Im-3m | 0.3113 | -1.1059 | 0.0139 | 0.0005 |
| $Mo_3Se_4$* | mp-21021 | R-3 | 0.4924 | -2.4124 | 0.0151 | 0.0004 |

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| $N_2$ | mp-1059834 | Imma | 1.3533 | -15.9343 | 0.0159 | 0.0010 |
| Na* | mp-10172 | P6_3/mmc | 0.0470 | -0.0839 | 0.1168 | 0.0901 |
| $Na_2Cl$ | mp-1077015 | Cmmm | 0.0377 | -0.1951 | 0.0901 | 0.0029 |
| $Na_2Cl$ | mp-990084 | R-3m | 0.0479 | -0.3527 | 0.0618 | 0.0000 |
| $Na_3Cl$ | mp-1064484 | P4/mmm | 0.0297 | -0.2254 | 0.0859 | 0.0249 |
| Nb* | mp-75 | Im-3m | 0.3268 | -1.1439 | 0.0000 | 0.0000 |
| $Nb_3Ir$* | mp-1458 | Pm-3n | 0.4144 | -2.0103 | 0.0000 | 0.0002 |
| $NbGa_3$* | mp-1973 | I4/mmm | 0.3104 | -1.7262 | 0.0000 | 0.0000 |
| NbO* | mp-2692 | Fm-3m | 0.2933 | -2.2812 | 0.0292 | 0.0023 |
| NbO | mp-634965 | P4/mmm | 0.9184 | -5.5839 | 0.0000 | 0.0000 |
| NbP* | mp-9830 | I4_1/amd | 0.3299 | -1.7803 | 0.0001 | 0.0000 |
| NbP | mp-1220317 | Cmmm | 0.3238 | -1.7805 | 0.0084 | 0.0004 |
| $NbRh_2$ | mp-1220414 | Pm | 0.3585 | -1.4459 | 0.0440 | 0.0023 |
| $NbS_2$ | mp-774873 | P6_3/mmc | 0.6390 | -3.1946 | 0.0049 | 0.0000 |
| Ni* | mp-23 | Fm-3m | 0.0947 | -0.4204 | 0.3799 | 0.2037 |
| NiB* | mp-14019 | Cmcm | 0.6996 | -6.1857 | 0.0198 | 0.0093 |
| $NiHg_4$* | mp-570835 | Im-3m | 0.3125 | -2.2179 | 0.0000 | 0.0004 |
| NiS* | mp-1547 | R3m | 0.2630 | -1.5895 | 0.1140 | 0.0342 |
| Os* | mp-49 | P6_3/mmc | 0.2567 | -1.1579 | 0.0538 | 0.0135 |
| OsC* | mp-7142 | P-6m2 | 0.4688 | -3.0941 | 0.0402 | 0.0068 |
| OsC | mp-1009537 | Pm-3m | 0.3353 | -2.0943 | 0.0446 | 0.0000 |
| OsC | mp-1009539 | Fm-3m | 0.4618 | -3.1760 | 0.0828 | 0.0058 |
| OsC | mp-999308 | P6_3/mmc | 0.4521 | -3.1583 | 0.0525 | 0.0226 |
| Pb* | mp-20483 | Fm-3m | 0.1943 | -0.8519 | 0.0276 | 0.0000 |
| $PbAu_2$* | mp-19871 | Fd-3m | 0.3048 | -2.3586 | 0.0000 | 0.0000 |
| Pd* | mp-2 | Fm-3m | 0.1631 | -1.2162 | 0.1264 | 0.0877 |
| Pt* | mp-126 | Fm-3m | 0.2367 | -1.8089 | 0.0927 | 0.0819 |
| Rb* | mp-70 | Im-3m | 0.0590 | -0.1180 | 0.1307 | 0.0377 |
| $Rb_2Hg_7$* | mp-31474 | P-3m1 | 0.1920 | -0.9499 | 0.0000 | 0.0000 |
| RbP* | mp-1179688 | C2/m | 0.1031 | -0.4288 | 0.0138 | 0.0467 |
| RbP | mp-1058499 | Immm | 0.1277 | -0.4304 | 0.0099 | 0.0324 |
| Re* | mp-8 | P6_3/mmc | 0.3162 | -1.3803 | 0.0205 | 0.0030 |
| $Re_3Ge_7$* | mp-29141 | Cmcm | 0.4553 | -2.2942 | 0.0022 | 0.0002 |
| ReC* | mp-1079494 | P6_3/mmc | 0.4741 | -3.3056 | 0.0211 | 0.0022 |
| ReC | mp-1009731 | Pm-3m | 0.3588 | -2.2826 | 0.0149 | 0.0003 |
| Rh* | mp-74 | Fm-3m | 0.1620 | -0.7081 | 0.1852 | 0.0924 |
| Ru* | mp-33 | P6_3/mmc | 0.2375 | -0.9478 | 0.0875 | 0.0211 |
| RuC* | mp-10030 | P-6m2 | 0.4770 | -2.9147 | 0.0418 | 0.0078 |
| RuC | mp-1009787 | Fm-3m | 0.4752 | -2.9734 | 0.0862 | 0.0067 |
| SN | mp-1078816 | P2_1/c | 0.2823 | -1.7903 | 0.0359 | 0.0006 |
| SN | mp-1173453 | P2_1 | 1.5195 | -10.9244 | 0.0458 | 0.0000 |
| SN | mp-684642 | P2_1 | 1.3504 | -13.4315 | 0.0333 | 0.0038 |
| Sb* | mp-104 | R-3m | 0.4398 | -1.7202 | 0.0425 | 0.0063 |
| Sc* | mp-67 | P6_3/mmc | 0.2110 | -0.6596 | 0.0000 | 0.0000 |
| $Sc_5Ge_3$* | mp-17190 | P6_3/mcm | 0.3162 | -1.5145 | 0.0000 | 0.0000 |
| $ScAg_2$* | mp-1018128 | I4/mmm | 0.2014 | -1.5516 | 0.0000 | 0.0000 |
| $ScTc_3$ | mp-867262 | Fm-3m | 0.3043 | -1.0739 | 0.0080 | 0.0010 |
| ScTe* | mp-10026 | P6_3/mmc | 0.3365 | -1.5673 | 0.0000 | 0.0000 |
| $SiAu_3$ | mp-1186998 | Fm-3m | 0.1974 | -1.4121 | 0.0156 | 0.0000 |
| $SiO_2$ | mp-556044 | P-31c | 0.7437 | -6.4399 | 0.0117 | 0.0000 |
| $SiO_2$ | mp-638049 | F4_132 | 0.6594 | -6.0556 | 0.0488 | 0.0000 |
| SiRh* | mp-2287 | P2_1/c | 0.2801 | -1.6004 | 0.0243 | 0.0009 |
| Sn* | mp-623511 | I4/mmm | 0.1989 | -0.8803 | 0.0235 | 0.0006 |
| $Sn_7Ir_5$* | mp-20466 | I4mm | 0.5375 | -3.2800 | 0.0000 | 0.0000 |
| $Sn_7Pd_5$ | mp-1219006 | C2 | 0.3765 | -2.7926 | 0.0000 | 0.0004 |
| SnPt* | mp-19856 | P6_3/mmc | 0.4612 | -3.2535 | 0.0000 | 0.0000 |

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| Sr* | mp-139 | P6_3/mmc | 0.0009 | -0.0171 | 4.9050 | 5.3659 |
| $Sr_2As$* | mp-15697 | I4/mmm | 0.0787 | -0.2746 | 0.2097 | 0.0267 |
| $Sr_3Sn$ | mp-1187136 | Pm-3m | 0.0501 | -0.2374 | 0.4377 | 0.0368 |
| $Sr_5Sn_3$* | mp-17720 | I4/mcm | 0.8182 | -2.1022 | 0.1796 | 0.0000 |
| $SrAl_2$* | mp-1638 | Fd-3m | 0.4263 | -1.6649 | 0.0004 | 0.0000 |
| SrN* | mp-1058586 | Fm-3m | 0.1230 | -0.6070 | 0.0462 | 0.0849 |
| SrSi* | mp-2661 | Cmcm | 0.8077 | -2.8883 | 0.0564 | 0.0097 |
| $SrTl_3$ | mp-1206636 | Pm-3m | 0.2857 | -1.2068 | 0.0201 | 0.0007 |
| Ta* | mp-50 | Im-3m | 0.3226 | -1.2794 | 0.0000 | 0.0000 |
| $Ta_4C_3$ | mp-1218000 | R3m | 0.5265 | -3.5440 | 0.0000 | 0.0000 |
| $Ta_5Ge_3$* | mp-17593 | I4/mcm | 0.3556 | -2.6739 | 0.0000 | 0.0000 |
| $Ta_7Co_6$* | mp-1104548 | R-3m | 0.4891 | -3.0211 | 0.0018 | 0.0000 |
| TaN* | mp-1497 | P6/mmm | 0.4241 | -2.9806 | 0.0009 | 0.0000 |
| TaN | mp-1009831 | Pm-3m | 0.3185 | -2.1377 | 0.0002 | 0.0002 |
| TaN | mp-1009833 | F-43m | 0.6451 | -4.4476 | 0.0000 | 0.0006 |
| TaP* | mp-1067587 | I4_1md | 0.4762 | -2.7480 | 0.0000 | 0.0000 |
| TaP | mp-1187244 | P-6m2 | 0.4792 | -2.7509 | 0.0000 | 0.0000 |
| TaRe | mp-1217894 | Cmmm | 0.3311 | -1.5114 | 0.0026 | 0.0003 |
| $TaSb_2$* | mp-11697 | C2/m | 0.4899 | -2.0334 | 0.0002 | 0.0000 |
| Tc* | mp-113 | P6_3/mmc | 0.3130 | -1.2061 | 0.0240 | 0.0006 |
| TcW | mp-1217337 | Cmmm | 0.6035 | -3.6836 | 0.0000 | 0.0002 |
| $Te_2Au$* | mp-567525 | C2/m | 0.5036 | -3.0627 | 0.0497 | 0.0327 |
| $Te_2Ir$* | mp-1551 | C2/m | 0.7621 | -4.4689 | 0.0230 | 0.0046 |
| $Te_2Pd$* | mp-782 | P-3m1 | 0.5813 | -3.5060 | 0.0291 | 0.0200 |
| Ti* | mp-46 | P6_3/mmc | 0.3010 | -0.7536 | 0.0000 | 0.0000 |
| $Ti_3Ge$* | mp-1189044 | I-4 | 0.3243 | -1.6051 | 0.0000 | 0.0000 |
| $Ti_3In$* | mp-866046 | P6_3/mmc | 0.3128 | -1.1764 | 0.0000 | 0.0000 |
| $Ti_3Pt$* | mp-2339 | Pm-3n | 0.3632 | -1.7698 | 0.0000 | 0.0000 |
| $TiBe_{12}$* | mp-1104067 | I4/mmm | 0.2186 | -1.9904 | 0.0000 | 0.0000 |
| $TiMo_2$ | mp-1216675 | Fmmm | 0.3395 | -1.1740 | 0.0000 | 0.0000 |
| TiNi* | mp-1048 | P2_1/m | 0.2818 | -0.8685 | 0.0059 | 0.0000 |
| TiNi | mp-1067248 | Cmcm | 0.2798 | -0.8586 | 0.0060 | 0.0000 |
| TiO* | mp-1071163 | P-62m | 0.3594 | -2.4547 | 0.0015 | 0.0000 |
| $TiSi_2$* | mp-1077503 | Cmcm | 0.4599 | -3.0700 | 0.0000 | 0.0000 |
| $TiSi_2$ | mp-570991 | Immm | 0.2659 | -1.2939 | 0.0000 | 0.0000 |
| TiW | mp-1216621 | Cmmm | 0.6568 | -3.6207 | 0.0000 | 0.0000 |
| Tl* | mp-82 | P6_3/mmc | 0.1905 | -1.0463 | 0.0000 | 0.0000 |
| $Tl_3Ga$ | mp-1187539 | Fm-3m | 0.2090 | -1.0914 | 0.0000 | 0.0000 |
| $Tl_4Sn$ | mp-1216560 | R-3m | 0.2039 | -1.3837 | 0.0010 | 0.0008 |
| TlC* | mp-567465 | Fm-3m | 0.3776 | -2.4265 | 0.0000 | 0.0010 |
| TlC | mp-1058887 | P4/mmm | 0.4837 | -3.1370 | 0.0000 | 0.0003 |
| TlN* | mp-1017982 | P6_3mc | 0.6066 | -3.6824 | 0.0023 | 0.0000 |
| TlN | mp-1002222 | Fm-3m | 0.3446 | -2.1516 | 0.0029 | 0.0000 |
| TlN | mp-1007778 | F-43m | 0.6105 | -3.6815 | 0.0017 | 0.0000 |
| V* | mp-146 | Im-3m | 0.3178 | -0.8777 | 0.0000 | 0.0000 |
| $VCo_3$ | mp-1095057 | P6_3/mmc | 0.3730 | -2.5141 | 0.0012 | 0.0023 |
| VN | mp-1017532 | P6_3/mmc | 0.4293 | -2.6954 | 0.0000 | 0.0000 |
| $VNi_2$* | mp-11531 | Immm | 0.4769 | -1.4029 | 0.0918 | 0.0008 |
| W* | mp-91 | Im-3m | 0.4372 | -2.5600 | 0.0000 | 0.0001 |
| WC* | mp-13136 | Fm-3m | 0.5508 | -4.2240 | 0.0012 | 0.0009 |
| WC | mp-1008630 | Pm-3m | 0.4127 | -2.9246 | 0.0000 | 0.0010 |
| WC | mp-1008635 | F-43m | 0.8710 | -6.2790 | 0.0000 | 0.0000 |
| $WO_2$* | mp-1524394 | P2_1/c | 0.6713 | -5.7408 | 0.0039 | 0.0000 |
| Y* | mp-112 | P6_3/mmc | 0.2374 | -0.8682 | 0.0000 | 0.0000 |
| $Y_3Co$ | mp-1189329 | P2_1 | 0.4718 | -2.6045 | 0.0000 | 0.0000 |
| $YAg_3$ | mp-1187811 | Fm-3m | 0.1780 | -1.2749 | 0.0000 | 0.0000 |

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| Zn* | mp-79 | P6_3/mmc | 0.2296 | -1.8892 | 0.0000 | 0.0000 |
| Zn | mp-2647117 | Im-3m | 0.1819 | -1.3641 | 0.0000 | 0.0000 |
| ZnAg* | mp-1912 | Pm-3m | 0.2035 | -1.8265 | 0.0000 | 0.0000 |
| $ZnPb_3$ | mp-1187949 | Fm-3m | 0.2392 | -1.3227 | 0.0004 | 0.0004 |
| Zr* | mp-131 | P6_3/mmc | 0.3223 | -1.3170 | 0.0000 | 0.0000 |
| $Zr_2Au$* | mp-2348 | I4/mmm | 0.3095 | -2.0203 | 0.0000 | 0.0000 |
| $Zr_3Be$ | mp-1188016 | I4/mmm | 0.3655 | -1.7390 | 0.0000 | 0.0000 |
| $Zr_3Ge$ | mp-1188039 | P6_3/mmc | 0.3460 | -1.7196 | 0.0000 | 0.0000 |
| $Zr_5Pb_3$* | mp-681992 | P6_3/mcm | 0.3301 | -1.9739 | 0.0000 | 0.0000 |
| ZrBr* | mp-1065846 | C2/m | 0.3528 | -2.0185 | 0.0000 | 0.0001 |
| $ZrGa_3$* | mp-30685 | I4/mmm | 0.2933 | -1.8532 | 0.0000 | 0.0000 |
| $ZrMo_2$* | mp-2049 | Fd-3m | 0.4671 | -1.6863 | 0.0002 | 0.0000 |
| ZrRh* | mp-2808 | Pm-3m | 0.3062 | -1.4732 | 0.0094 | 0.0024 |
| ZrTi | mp-1215236 | P4/mmm | 0.3777 | -1.6592 | 0.0000 | 0.0000 |

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

## A.2 Ionic

Table S2: Ionic compounds (non-metallic, ICOBI $<$ 0.5052, observed range 0.0444 to 0.4976) — same columns.

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| AgO* | mp-1079720 | P2_1/c | 0.2604 | -2.0408 | 0.0297 | 0.0024 |
| $B_6Pb$ | mp-1096825 | Pm-3m | 0.2353 | -1.9376 | 0.0000 | 0.0000 |
| $BaCl_2$* | mp-568662 | Fm-3m | 0.0749 | -0.5008 | 0.2998 | 0.0066 |
| $BaCl_2$* | mp-23199 | Pnma | 0.0637 | -0.4308 | 0.3027 | 0.0019 |
| $BaF_3$* | mp-1080821 | P2_1/m | 0.2279 | -0.7883 | 0.3169 | 0.0437 |
| $BaH_2$* | mp-23715 | Pnma | 0.0615 | -0.2536 | 0.1495 | 0.0000 |
| BaO* | mp-1008500 | P6_3/mmc | 0.0788 | -0.6380 | 0.1468 | 0.0016 |
| BaO | mp-1950989 | P6_3/mmc | 0.1070 | -0.4367 | 0.2225 | 0.0000 |
| $Be_3N_2$* | mp-6977 | P6_3/mmc | 0.4361 | -4.2580 | 0.0000 | 0.0000 |
| $Be_3N_2$* | mp-18337 | Ia-3 | 0.4759 | -4.7015 | 0.0000 | 0.0000 |
| $Be_3N_2$ | mp-1017550 | P-3m1 | 0.3952 | -3.8753 | 0.0000 | 0.0000 |
| $BeCl_2$ | mp-1172909 | I-43m | 0.3697 | -0.9454 | 0.8095 | 0.1761 |
| $Ca_3N_2$* | mp-1047 | R-3c | 0.1781 | -0.6015 | 0.1048 | 0.0000 |
| $Ca_3N_2$* | mp-844 | Ia-3 | 0.1820 | -0.6211 | 0.0837 | 0.0000 |
| $CaCl_2$* | mp-23214 | Pnnm | 0.1054 | -0.7540 | 0.2048 | 0.0008 |
| $CaCl_2$ | mp-1120739 | Fm-3m | 0.0814 | -0.5708 | 0.3236 | 0.0068 |
| $CaF_2$* | mp-10464 | Pnma | 0.0565 | -0.5566 | 0.2424 | 0.0000 |
| $CaF_2$ | mp-554355 | P4/mmm | 0.0591 | -0.5440 | 0.1286 | 0.0000 |
| CaO* | mp-1079707 | Cmce | 0.1008 | -0.5793 | 0.1805 | 0.0000 |
| CaO* | mp-2605 | Fm-3m | 0.1143 | -0.7028 | 0.0897 | 0.0000 |
| $CdI_2$* | mp-567259 | P-3m1 | 0.4859 | -2.4996 | 0.0508 | 0.0000 |
| $Cs_2S$* | mp-1077814 | P6_3/mmc | 0.0804 | -1.0011 | 0.0682 | 0.0000 |
| $Cs_2Se$* | mp-1011696 | Pnma | 0.0811 | -0.6189 | 0.0861 | 0.0000 |
| CsCl* | mp-573697 | Fm-3m | 0.0595 | -0.3453 | 0.0490 | 0.0000 |
| CsF* | mp-8455 | Pm-3m | 0.1224 | -1.5102 | 0.0523 | 0.0008 |
| CsH* | mp-632319 | Pm-3m | 0.0935 | -1.2305 | 0.0678 | 0.0002 |
| GeS | mp-12910 | Cmcm | 0.4348 | -2.2462 | 0.0516 | 0.0000 |
| $H_2O$* | mp-2739191 | C222_1 | 0.4372 | -4.6426 | 0.0087 | 0.0000 |
| HI* | mp-697025 | C2/c | 0.4915 | -2.7099 | 0.0522 | 0.0000 |
| $HgF_2$* | mp-8177 | Fm-3m | 0.2320 | -1.8744 | 0.1660 | 0.0000 |
| I* | mp-23153 | Cmce | 0.0444 | -0.1504 | 0.4981 | 0.0900 |

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| InBr* | mp-22870 | Cmcm | 0.3010 | -1.3292 | 0.0929 | 0.0024 |
| InCl* | mp-571636 | Cmc2_1 | 0.1698 | -0.8368 | 0.0921 | 0.0006 |
| $K_2O$* | mp-971 | Fm-3m | 0.1009 | -0.8814 | 0.0059 | 0.0000 |
| $K_2S$* | mp-559278 | P6_3/mmc | 0.1040 | -0.4891 | 0.0071 | 0.0006 |
| $K_2S$* | mp-1022 | Fm-3m | 0.1212 | -0.4744 | 0.0000 | 0.0000 |
| KCl* | mp-23289 | Pm-3m | 0.0510 | -0.4078 | 0.1374 | 0.0000 |
| $KF_3$* | mp-1544557 | P2_12_12_1 | 0.3162 | -0.9256 | 0.4749 | 0.3545 |
| $KF_3$ | mp-2749823 | P2_1 | 0.3119 | -0.9244 | 0.4733 | 0.3514 |
| KI* | mp-22898 | Fm-3m | 0.0893 | -0.4549 | 0.0615 | 0.0000 |
| $Li_2O$* | mp-1960 | Fm-3m | 0.2393 | -2.0320 | 0.0000 | 0.0000 |
| $Li_2O$ | mp-755894 | Pnma | 0.2225 | -2.0369 | 0.0000 | 0.0000 |
| $Li_2S$* | mp-1125 | Pnma | 0.2847 | -2.0194 | 0.0000 | 0.0000 |
| $Li_2S$ | mp-557142 | P6_3/mmc | 0.3327 | -2.1900 | 0.0000 | 0.0000 |
| $Li_2Se$* | mp-2286 | Fm-3m | 0.3399 | -2.1702 | 0.0000 | 0.0000 |
| $Li_3N$* | mp-2341 | P6_3/mmc | 0.2587 | -1.8972 | 0.0000 | 0.0000 |
| LiBr | – | – | 0.2790 | -2.0381 | 0.0000 | 0.0000 |
| LiCl* | mp-22905 | Fm-3m | 0.2558 | -2.0554 | 0.0000 | 0.0000 |
| LiCl | mp-1185319 | P6_3mc | 0.3879 | -2.9688 | 0.0000 | 0.0000 |
| LiF* | mp-1138 | Fm-3m | 0.1692 | -1.9334 | 0.0001 | 0.0000 |
| LiF | mp-1009009 | Pm-3m | 0.1237 | -1.3411 | 0.0249 | 0.0000 |
| $MgCl_2$* | mp-570259 | P-3m1 | 0.1312 | -0.9334 | 0.2905 | 0.0028 |
| $MgCl_2$* | mp-23210 | R-3m | 0.1313 | -0.9344 | 0.2883 | 0.0009 |
| $MgCl_2$ | mp-569626 | Ama2 | 0.1475 | -1.0096 | 0.1898 | 0.0001 |
| $MgF_2$* | mp-1072956 | Pnnm | 0.0901 | -1.0192 | 0.1330 | 0.0000 |
| $MgF_2$ | mp-1062842 | Amm2 | 0.0808 | -0.8465 | 0.1522 | 0.0000 |
| $MgF_2$ | mp-560236 | C2/m | 0.1007 | -1.0813 | 0.0878 | 0.0000 |
| $MgH_2$* | mp-23711 | Pbcn | 0.1156 | -0.5595 | 0.0617 | 0.0000 |
| MgO* | mp-549706 | P6_3mc | 0.1572 | -1.2025 | 0.0812 | 0.0000 |
| MgO | mp-1009127 | Pm-3m | 0.0813 | -0.6164 | 0.2735 | 0.0000 |
| MgS* | mp-1315 | Fm-3m | 0.1236 | -0.7426 | 0.2438 | 0.0000 |
| MgS | mp-13032 | F-43m | 0.2006 | -1.1035 | 0.1624 | 0.0000 |
| $Na_2O$* | mp-2352 | Fm-3m | 0.1191 | -0.6209 | 0.0000 | 0.0000 |
| $Na_2O$ | mp-1975297 | C2/m | 0.1297 | -0.6320 | 0.0003 | 0.0000 |
| NaCl | – | – | 0.0859 | -0.5953 | 0.1111 | 0.0000 |
| NaS* | mp-409 | P-62m | 0.4976 | -2.6241 | 0.1126 | 0.0000 |
| NaS | mp-1180106 | C2/m | 0.2104 | -0.9839 | 0.0000 | 0.0000 |
| PbSe* | mp-22009 | Fmm2 | 0.4169 | -1.6445 | 0.0576 | 0.0000 |
| PbSe* | mp-2201 | Fm-3m | 0.3457 | -1.4608 | 0.0807 | 0.0039 |
| $Rb_2O$* | mp-1394 | Fm-3m | 0.0867 | -1.2998 | 0.0368 | 0.0000 |
| $Rb_2O$ | mp-2023482 | P-3m1 | 0.1302 | -0.4067 | 0.0636 | 0.0000 |
| RbCl* | mp-23299 | Pm-3m | 0.0478 | -0.6732 | 0.1019 | 0.0000 |
| RbF* | mp-2064 | Pm-3m | 0.0879 | -1.3700 | 0.0447 | 0.0036 |
| RbF | mp-11718 | Fm-3m | 0.0460 | -0.4536 | 0.0398 | 0.0000 |
| RbI* | mp-22903 | Fm-3m | 0.0840 | -0.4162 | 0.0534 | 0.0013 |
| RbS | mp-1179684 | C2/m | 0.1226 | -0.5861 | 0.0850 | 0.0092 |
| $Sc_2O_3$ | mp-755313 | R-3c | 0.4210 | -2.7698 | 0.0026 | 0.0000 |
| $SiO_2$ | mp-10064 | Fm-3m | 0.3974 | -3.7939 | 0.0240 | 0.0000 |
| $SnBr_2$ | mp-978985 | P4_2/mnm | 0.3595 | -1.6332 | 0.1289 | 0.0000 |
| $SrF_2$* | mp-1102240 | Pnma | 0.0589 | -0.5401 | 0.2090 | 0.0000 |
| $SrF_2$ | mp-1102911 | Pnma | 0.0505 | -0.4858 | 0.2166 | 0.0000 |
| SrS* | mp-10627 | Pm-3m | 0.0762 | -0.4641 | 0.2376 | 0.0000 |
| Te* | mp-19 | P3_121 | 0.3172 | -1.1233 | 0.1197 | 0.0127 |
| $Te_2Ru$* | mp-267 | Pnnm | 0.4723 | -2.2556 | 0.0486 | 0.0017 |
| TlCl* | mp-571079 | Cmcm | 0.1717 | -0.8409 | 0.1079 | 0.0000 |

| | | | | | | |
|---|---|---|---|---|---|---|
| $Y_2O_3$* | mp-558573 | C2/m | 0.4214 | -2.9843 | 0.0000 | 0.0000 |

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

## A.3 Covalent

Table S3: Covalent compounds (non-metallic, ICOBI $\geq$ 0.5052, observed range 0.5087 to 1.9179) — same columns.

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| $As_2S_3$* | mp-1189572 | P-1 | 0.9136 | -4.7416 | 0.0538 | 0.0000 |
| $AsF_3$* | mp-28027 | Pna2_1 | 0.6951 | -6.3410 | 0.1114 | 0.0000 |
| B* | mp-160 | R-3m | 0.5203 | -4.7960 | 0.0000 | 0.0000 |
| $BI_3$* | mp-23189 | P6_3/m | 1.2727 | -6.8061 | 0.0232 | 0.0000 |
| $Be_3P_2$* | mp-567841 | Ia-3 | 0.5637 | -3.8282 | 0.0000 | 0.0000 |
| $Be_3P_2$ | mp-1084771 | Pn-3m | 0.5501 | -3.7626 | 0.0000 | 0.0002 |
| $BeCl_2$* | mp-23267 | Ibam | 0.7165 | -5.6352 | 0.0021 | 0.0000 |
| $BeCl_2$ | mp-1190080 | I-43m | 0.7528 | -5.9613 | 0.0026 | 0.0000 |
| $BeF_2$* | mp-15951 | P3_121 | 0.5269 | -6.8585 | 0.0035 | 0.0000 |
| BeO* | mp-7599 | P4_2/mnm | 0.5148 | -5.7846 | 0.0000 | 0.0000 |
| BeO | mp-1778 | F-43m | 0.5232 | -5.8507 | 0.0000 | 0.0000 |
| $Bi_2O_3$ | mp-714859 | P1 | 1.1110 | -10.1326 | 0.0646 | 0.0001 |
| Br* | mp-23154 | Cmce | 0.7827 | -3.5362 | 0.4257 | 0.0097 |
| C | mp-1080826 | C2/m | 0.9446 | -9.5768 | 0.0000 | 0.0000 |
| C | mp-1190171 | Pnma | 0.9445 | -9.5798 | 0.0000 | 0.0000 |
| C* | mp-66 | Fd-3m | 0.9497 | -9.6602 | 0.0000 | 0.0000 |
| C* | mp-47 | P6_3/mmc | 0.9482 | -9.7832 | 0.0000 | 0.0000 |
| C | – | – | 1.2173 | -11.6891 | 0.0000 | 0.0000 |
| $C_3N_4$* | mp-571653 | P-43m | 0.9531 | -10.4956 | 0.0150 | 0.0000 |
| $C_3N_4$ | mp-1182484 | P6_3/m | 1.9179 | -15.3586 | 0.0000 | 0.0000 |
| $C_3N_4$ | mp-2852 | I-43d | 0.9575 | -10.9099 | 0.0106 | 0.0000 |
| $C_3N_4$ | mp-563 | Fd-3m | 0.7047 | -7.1477 | 0.0086 | 0.0000 |
| $Cl_2$* | mp-22848 | Cmce | 0.9254 | -5.5728 | 0.4555 | 0.0003 |
| $ClF_3$* | mp-556767 | Pnma | 0.6390 | -5.2173 | 0.1347 | 0.0000 |
| $F_2$* | mp-561203 | C2/c | 0.9731 | -6.9536 | 0.4569 | 0.0000 |
| $Ga_2Se_3$ | mp-1224809 | Imm2 | 0.8324 | -4.7105 | 0.0085 | 0.0000 |
| $GeSe_2$* | mp-10074 | I-42d | 0.9011 | -4.9682 | 0.0171 | 0.0000 |
| $H_2$* | mp-730101 | P2_12_12_1 | 0.9954 | -5.1222 | 0.0000 | 0.0000 |
| $H_3N$* | mp-643432 | P2_12_12_1 | 0.8816 | -8.0639 | 0.0000 | 0.0000 |
| HCl | mp-684609 | Imm2 | 0.9480 | -7.3985 | 0.0478 | 0.0001 |
| $IF_7$* | mp-27988 | Aea2 | 0.5305 | -5.5662 | 0.0568 | 0.0000 |
| $IrCl_3$ | mp-729507 | C2 | 0.7112 | -4.9995 | 0.0957 | 0.0000 |
| $Mn_2O_7$* | mp-28338 | P2_1/c | 1.3609 | -6.7339 | 0.0257 | 0.0000 |
| MnN | mp-1009130 | F-43m | 0.6682 | -3.9792 | 0.0247 | 0.0062 |
| $MoSe_2$* | mp-1634 | P6_3/mmc | 0.6411 | -2.8467 | 0.0035 | 0.0000 |
| $Nb_2O_5$* | mp-1595 | C2/m | 0.7548 | -4.9296 | 0.0075 | 0.0000 |
| $O_2$* | mp-1524462 | C2/m | 1.4816 | -17.7956 | 0.1271 | 0.2186 |
| P* | mp-568348 | P2/c | 0.9202 | -5.0760 | 0.0209 | 0.0000 |
| PS* | mp-612 | C2/c | 0.9840 | -5.8992 | 0.0301 | 0.0000 |
| $PtCl_2$ | mp-1219748 | C2/m | 0.7183 | -5.3013 | 0.0756 | 0.0000 |
| S* | mp-77 | Fddd | 0.9692 | -5.9246 | 0.0880 | 0.0000 |
| SN* | mp-236 | P2_1/c | 1.1480 | -8.7413 | 0.0440 | 0.0000 |
| Se* | mp-14 | P3_121 | 0.9053 | -4.1496 | 0.0828 | 0.0000 |
| Si | – | – | 0.9243 | -4.5418 | 0.0000 | 0.0000 |
| $SiO_2$* | mp-546794 | I-42d | 0.7466 | -7.9383 | 0.0071 | 0.0000 |

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| $SiO_2$* | mp-9258 | Pa-3 | 0.5130 | -5.5969 | 0.0264 | 0.0000 |
| $SiS_2$* | mp-1095294 | P2_1/c | 0.9089 | -5.7585 | 0.0044 | 0.0000 |
| $TeO_2$* | mp-8377 | P2_12_12_1 | 0.6215 | -4.7653 | 0.0772 | 0.0000 |
| $TiO_2$* | mp-1439 | Pbcn | 0.5838 | -3.5796 | 0.0077 | 0.0000 |
| $TiO_2$* | mp-9173 | Pnma | 0.5897 | -3.5580 | 0.0123 | 0.0000 |
| $TiO_2$* | mp-430 | P2_1/c | 0.5087 | -3.0819 | 0.0145 | 0.0000 |
| $ZrO_2$ | mp-775909 | P4_2/mnm | 0.5720 | -4.1446 | 0.0017 | 0.0000 |

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

## A.4 Mixed (Zintl-type)

Table S4: Mixed-bonding (Zintl-type) compounds (a strongly covalent homoatomic pair coexists with a markedly weaker heteroatomic pair, e.g. the azide $N_3^-$ ion ionically bound to its cation; observed range 0.8522 to 2.7322) — same columns.

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|
| $Ba_5P_4$* | mp-17566 | Pnma | 0.9040 | -3.4927 | 0.2104 | 0.0000 |
| $BaS_3$* | mp-556296 | P2_12_12 | 0.9201 | -5.2850 | 0.1084 | 0.0000 |
| $Bi_2O_3$ | mp-674644 | P1 | 0.9332 | -7.5456 | 0.0131 | 0.0000 |
| $CdO_2$* | mp-2310 | Pa-3 | 0.9738 | -7.1680 | 0.0540 | 0.0000 |
| $Cs_2As_3$* | mp-15557 | Fmmm | 1.0431 | -4.4248 | 0.0386 | 0.0000 |
| $CsN_3$* | mp-510557 | I4/mcm | 1.8888 | -17.7624 | 0.0000 | 0.0000 |
| $CsN_3$ | mp-1225892 | P4/mmm | 1.8983 | -17.5725 | 0.0000 | 0.0000 |
| $CsP_7$* | mp-1201790 | Pbcm | 0.9343 | -4.9186 | 0.0062 | 0.0000 |
| $CsP_7$ | mp-1213206 | Pca2_1 | 0.9336 | -4.9318 | 0.0064 | 0.0000 |
| $KN_3$* | mp-827 | I4/mcm | 1.8889 | -18.4973 | 0.0000 | 0.0000 |
| $LiP_5$* | mp-2412 | Pna2_1 | 0.9037 | -4.9376 | 0.0026 | 0.0000 |
| $LiP_5$ | mp-32760 | Pna2_1 | 0.8771 | -4.6811 | 0.0064 | 0.0000 |
| $MgP_4$* | mp-384 | P2_1/c | 0.9196 | -5.0216 | 0.0273 | 0.0000 |
| $NaN_3$* | mp-1066400 | R3m | 1.8853 | -18.8113 | 0.0000 | 0.0000 |
| $NaN_3$* | mp-22003 | R-3m | 1.8872 | -19.1626 | 0.0002 | 0.0000 |
| $NaN_3$ | mp-1065265 | C2/m | 2.7322 | -23.2128 | 0.0006 | 0.0215 |
| NaS* | mp-2400 | P6_3/mmc | 0.9510 | -4.7236 | 0.1279 | 0.0000 |
| $PI_2$* | mp-29443 | P-1 | 0.9586 | -4.8005 | 0.1081 | 0.0000 |
| $PbN_6$* | mp-667338 | Pnma | 1.8687 | -19.2227 | 0.0007 | 0.0000 |
| $RbN_3$* | mp-581833 | P4/mmm | 1.8969 | -18.3403 | 0.0000 | 0.0000 |
| $RbN_3$ | mp-1219567 | Amm2 | 1.3756 | -9.8572 | 0.0216 | 0.0000 |
| RbS* | mp-558071 | Immm | 0.9920 | -5.0554 | 0.1964 | 0.0019 |
| $Sr_3As_4$ | mp-678007 | P1 | 0.8522 | -3.0556 | 0.1161 | 0.0000 |
| $Sr_3P_4$* | mp-14288 | Fdd2 | 0.8910 | -4.1353 | 0.1224 | 0.0000 |
| $Sr_3P_4$ | mp-682953 | P1 | 0.8889 | -4.0705 | 0.1250 | 0.0000 |
| $SrO_2$* | mp-2697 | I4/mmm | 0.9835 | -7.0573 | 0.1841 | 0.0000 |
| $SrO_2$* | mp-726705 | Pnma | 0.9821 | -6.8700 | 0.2389 | 0.0000 |
| $SrO_2$ | mp-2763179 | Pnma | 0.9805 | -6.9738 | 0.2327 | 0.0000 |

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

## A.5 Unclassified

Table S5: Unclassified compounds (no ICOBI data available) — same columns, ICOBI/antibonding shown as "–" where missing.

| Formula | mp-id | Space group | ICOBI | ICOHP (eV) | ICOHP antibonding | ICOBI antibonding |
|---|---|---|---|---|---|---|

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

# B VASP Calculation Parameters

Fixed INCAR parameters, identical across the whole dataset (static, LOBSTER-compatible run) — only `NBANDS` (appendix ??) and the k-point grid vary by compound.

| Parameter | Value | Description |
|---|---|---|
| `PREC` | Accurate | Overall precision (charge grid, projectors) |
| `ENCUT` | 600.0 | Plane-wave cutoff energy (eV) |
| `EDIFF` | 1e-06 | Electronic convergence criterion (eV) |
| `IBRION` | -1 | Ionic dynamics type ($-1$ = static, no relaxation) |
| `NSW` | 0 | Number of ionic steps (0 = static run on the MP structure) |
| `ISMEAR` | 0 | Smearing type (0 = Gaussian) |
| `SIGMA` | 0.05 | Smearing width (eV) |
| `LASPH` | True | Non-spherical corrections to the PAW potential |
| `ISYM` | -1 | Symmetry ($-1$ = disabled, required by LOBSTER) |
| `LWAVE` | True | Write WAVECAR (required by LOBSTER) |
| `LCHARG` | True | Write CHGCAR |
| `NPAR` | 4 | Band-level parallelization |
| `KPAR` | 2 | K-point-level parallelization |

# C PAW-PBE Potentials

One PAW-PBE potential per element present in the dataset (pymatgen's `MPRelaxSet` library — the same one Materials Project used to relax the input structures), with one documented exception for tungsten (§??): the recommended `W_pv` variant is absent from the local mirror, and `W_sv` turned out to be corrupted (missing `LEXCH` field) — plain `W` is used instead, verified valid.

Table S6: PAW-PBE potentials used, one per element present in the dataset (pymatgen's MPRelaxSet library, with one documented exception: W substituted for W_pv/W_sv, see §7).

| Element | PAW-PBE potential |
|---|---|
| Ag | `Ag (02Apr2005)` |
| Al | `Al (04Jan2001)` |
| As | `As (22Sep2009)` |
| Au | `Au (04Oct2007)` |
| B | `B (06Sep2000)` |
| Ba | `Ba_sv (06Sep2000)` |

| Element | PAW-PBE potential |
|---|---|
| Be | Be_sv (06Sep2000) |
| Bi | Bi (08Apr2002) |
| Br | Br (06Sep2000) |
| C | C (08Apr2002) |
| Ca | Ca_sv (06Sep2000) |
| Cd | Cd (06Sep2000) |
| Cl | Cl (06Sep2000) |
| Co | Co (02Aug2007) |
| Cr | Cr_pv (02Aug2007) |
| Cs | Cs_sv (25Jan2019) |
| Cu | Cu_pv (06Sep2000) |
| F | F (08Apr2002) |
| Fe | Fe_pv (02Aug2007) |
| Ga | Ga_d (06Jul2010) |
| Ge | Ge_d (03Jul2007) |
| H | H (15Jun2001) |
| Hf | Hf_pv (06Sep2000) |
| Hg | Hg (06Sep2000) |
| I | I (08Apr2002) |
| In | In_d (06Sep2000) |
| Ir | Ir (06Sep2000) |
| K | K_sv (06Sep2000) |
| Li | Li_sv (10Sep2004) |
| Mg | Mg_pv (13Apr2007) |
| Mn | Mn_pv (02Aug2007) |
| Mo | Mo_pv (04Feb2005) |
| N | N (08Apr2002) |
| Na | Na_pv (19Sep2006) |
| Nb | Nb_pv (08Apr2002) |
| Ni | Ni_pv (06Sep2000) |
| O | O (08Apr2002) |
| Os | Os_pv (20Jan2003) |
| P | P (06Sep2000) |
| Pb | Pb_d (06Sep2000) |
| Pd | Pd (04Jan2005) |
| Pt | Pt (04Feb2005) |
| Rb | Rb_sv (06Sep2000) |
| Re | Re_pv (06Sep2000) |
| Rh | Rh_pv (25Jan2005) |
| Ru | Ru_pv (28Jan2005) |
| S | S (06Sep2000) |
| Sb | Sb (06Sep2000) |
| Sc | Sc_sv (07Sep2000) |
| Se | Se (06Sep2000) |
| Si | Si (05Jan2001) |
| Sn | Sn_d (06Sep2000) |
| Sr | Sr_sv (07Sep2000) |
| Ta | Ta_pv (07Sep2000) |
| Tc | Tc_pv (04Feb2005) |
| Te | Te (08Apr2002) |
| Ti | Ti_pv (07Sep2000) |
| Tl | Tl_d (06Sep2000) |
| V | V_pv (07Sep2000) |
| W | W (08Apr2002) |
| Y | Y_sv (25May2007) |
| Zn | Zn (06Sep2000) |

| Element | PAW-PBE potential |
|---|---|
| Zr | `Zr_sv (04Jan2005)` |

# D k-point Grid and NBANDS per Compound

k-point grid auto-generated by `pymatgen.io.vasp.Kpoints.automatic_density_by_vol` (density `kppvol=100`, Γ-centered or Monkhorst–Pack mesh depending on cell symmetry) and `NBANDS`, derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`) — a strictly sufficient value for the local-basis projection, not a fixed guess.

Table S7: k-point grid (`pymatgen` automatic density, `kppvol=100`) and number of bands (derived from the LOBSTER basis) per compound.

| Formula | mp-id | k-point grid | NBANDS |
|---|---|---|---|
| **Experimental, stable (hull)** ($n = 63$) | | | |
| $As_2S_3$ | mp-1189572 | 5 3 2 (G) | 80 |
| $AsF_3$ | mp-28027 | 5 4 4 (G) | 64 |
| $BI_3$ | mp-23189 | 4 4 4 (G) | 32 |
| BW | mp-7832 | 9 9 3 (G) | 52 |
| $BaAu_2$ | mp-30363 | 6 6 7 (G) | 23 |
| $BaCl_2$ | mp-568662 | 6 6 6 (G) | 13 |
| $BaH_2$ | mp-23715 | 6 4 3 (G) | 28 |
| $Be_2Cu$ | mp-2031 | 7 7 7 (G) | 38 |
| $Be_2Nb_3$ | mp-11272 | 8 4 4 (M) | 74 |
| $Be_2V$ | mp-11281 | 7 7 4 (G) | 76 |
| $CaAl_4$ | mp-1749 | 7 7 4 (G) | 21 |
| CaSn | mp-2450 | 6 6 4 (M) | 28 |
| CdPd | mp-1696 | 7 6 6 (G) | 36 |
| CoB | mp-20857 | 9 7 5 (G) | 52 |
| $Cs_2As_3$ | mp-15557 | 4 4 3 (G) | 44 |
| FeS | mp-505531 | 8 8 5 (G) | 26 |
| $Ge_2Os$ | mp-10032 | 9 6 4 (G) | 54 |
| $HfTc_2$ | mp-1095669 | 5 5 3 (G) | 108 |
| $Hg_4Pt$ | mp-936 | 5 5 5 (G) | 45 |
| $HgF_2$ | mp-8177 | 8 8 8 (G) | 17 |
| InBr | mp-22870 | 6 6 4 (M) | 26 |
| KI | mp-22898 | 6 6 6 (G) | 9 |
| $KN_3$ | mp-827 | 5 5 5 (G) | 34 |
| $Li_2Se$ | mp-2286 | 7 7 7 (G) | 14 |
| LiB | mp-1001835 | 7 7 9 (G) | 18 |
| $Mg_2Ni$ | mp-2137 | 5 5 2 (G) | 102 |
| $MgP_4$ | mp-384 | 5 5 3 (G) | 40 |
| $MnAl_{12}$ | mp-1104165 | 4 4 4 (M) | 57 |
| $MoSe_2$ | mp-1634 | 9 9 2 (G) | 34 |
| NaS | mp-2400 | 6 6 3 (G) | 32 |
| Na | mp-10172 | 8 8 4 (G) | 8 |
| $Nb_3Ir$ | mp-1458 | 5 5 5 (G) | 72 |
| $NbGa_3$ | mp-1973 | 8 8 6 (M) | 36 |
| $PI_2$ | mp-29443 | 6 4 3 (G) | 24 |
| PbSe | mp-2201 | 7 7 7 (G) | 13 |
| RbI | mp-22903 | 6 6 6 (G) | 9 |
| $Sc_5Ge_3$ | mp-17190 | 3 3 5 (G) | 154 |
| $ScAg_2$ | mp-1018128 | 8 8 5 (G) | 28 |

Table S7 (continued)

| Formula | mp-id | k-point grid | NBANDS |
|---|---|---|---|
| ScTe | mp-10026 | 7 7 4 (G) | 28 |
| SnPt | mp-19856 | 7 7 5 (G) | 36 |
| $SrO_2$ | mp-2697 | 8 8 7 (G) | 13 |
| SrSi | mp-2661 | 7 6 4 (G) | 18 |
| $Ta_5Ge_3$ | mp-17593 | 4 4 4 (M) | 144 |
| TaP | mp-1067587 | 9 9 4 (G) | 26 |
| $TaSb_2$ | mp-11697 | 8 5 4 (G) | 34 |
| $Te_2Pd$ | mp-782 | 7 7 5 (G) | 17 |
| $Te_2Ru$ | mp-267 | 7 5 4 (G) | 34 |
| $TeO_2$ | mp-8377 | 6 5 3 (G) | 48 |
| $Ti_3In$ | mp-866046 | 5 5 6 (G) | 72 |
| $Ti_3Pt$ | mp-2339 | 5 5 5 (G) | 72 |
| $TiBe_{12}$ | mp-1104067 | 7 5 5 (G) | 69 |
| TiNi | mp-1048 | 10 7 6 (G) | 36 |
| TiO | mp-1071163 | 10 6 6 (G) | 39 |
| $TiSi_2$ | mp-1077503 | 8 8 4 (M) | 34 |
| TlCl | mp-571079 | 6 6 4 (M) | 26 |
| $VNi_2$ | mp-11531 | 12 8 7 (G) | 27 |
| $WO_2$ | mp-1524394 | 6 5 5 (G) | 68 |
| $Zr_2Au$ | mp-2348 | 9 9 4 (G) | 29 |
| $Zr_5Pb_3$ | mp-681992 | 3 3 5 (G) | 154 |
| $ZrMo_2$ | mp-2049 | 6 6 6 (G) | 56 |
| Si | mp-149 | 8 8 8 (G) | 100 |
| NaCl | mp-22862 | 8 8 8 (G) | 100 |
| AlNi | mp-1487 | 10 10 10 (M) | 100 |
| **Experimental, metastable** ($n = 63$) | | | |
| $CdI_2$ | mp-567259 | 7 7 4 (G) | 17 |
| $MgCl_2$ | mp-570259 | 8 8 5 (G) | 12 |
| C | mp-169 | 12 12 8 (M) | 100 |
| ZrBr | mp-1065846 | 8 8 3 (G) | 28 |
| $Li_3Hg$ | mp-1646 | 7 7 7 (G) | 24 |
| $Ta_7Co_6$ | mp-1104548 | 6 6 3 (G) | 117 |
| $CaGa_4$ | mp-2461 | 7 7 5 (G) | 41 |
| InCl | mp-571636 | 6 6 4 (M) | 26 |
| NiB | mp-14019 | 10 10 7 (G) | 26 |
| $MgH_2$ | mp-23711 | 6 5 5 (G) | 24 |
| $Sn_7Ir_5$ | mp-20466 | 4 4 4 (M) | 108 |
| $NaN_3$ | mp-1066400 | 7 7 7 (G) | 16 |
| $ClF_3$ | mp-556767 | 6 4 3 (G) | 64 |
| $SiS_2$ | mp-1095294 | 5 4 3 (G) | 48 |
| $Sr_5Sn_3$ | mp-17720 | 3 3 3 (G) | 104 |
| $Te_2Ir$ | mp-1551 | 7 5 2 (G) | 51 |
| $Li_{13}Sn_5$ | mp-30769 | 6 6 1 (G) | 110 |
| $H_3N$ | mp-643432 | 8 5 5 (G) | 28 |
| $Cs_2Se$ | mp-1011696 | 5 3 2 (G) | 56 |
| $In_3Ir$ | mp-630976 | 4 4 4 (M) | 144 |
| HI | mp-697025 | 3 3 3 (G) | 40 |
| $Rb_2Hg_7$ | mp-31474 | 4 4 4 (G) | 73 |
| $PbAu_2$ | mp-19871 | 5 5 5 (G) | 54 |
| BeCu | mp-2323 | 10 10 10 (M) | 100 |
| $SiO_2$ | mp-546794 | 6 6 6 (M) | 24 |
| $IF_7$ | mp-27988 | 5 5 3 (G) | 64 |
| K | mp-10157 | 6 6 6 (G) | 5 |
| $Al_3Os_2$ | mp-16521 | 9 9 3 (G) | 30 |

Table S7 (continued)

| Formula | mp-id | k-point grid | NBANDS |
|---|---|---|---|
| $Nb_2O_5$ | mp-1595 | 7 7 4 (G) | 38 |
| LiBr | mp-23259 | 8 8 8 (G) | 100 |
| ZnAg | mp-1912 | 9 9 9 (G) | 18 |
| $MgPd_3$ | mp-7753 | 7 7 7 (G) | 31 |
| $AsPd_2$ | mp-1080831 | 8 4 4 (G) | 66 |
| $GeSe_2$ | mp-10074 | 5 5 5 (G) | 34 |
| $Re_3Ge_7$ | mp-29141 | 9 6 1 (G) | 180 |
| $Cu_3Ge$ | mp-19724 | 6 6 5 (G) | 72 |
| $Ti_3Ge$ | mp-1189044 | 4 4 4 (M) | 144 |
| $ZrGa_3$ | mp-30685 | 7 7 5 (G) | 37 |
| $SrAl_2$ | mp-1638 | 5 5 5 (G) | 26 |
| NiS | mp-1547 | 6 6 6 (M) | 39 |
| $Al_{12}W$ | mp-11227 | 4 4 4 (M) | 57 |
| CsH | mp-632319 | 7 7 7 (G) | 6 |
| $CaGa_2$ | mp-992 | 7 7 6 (G) | 23 |
| $NiHg_4$ | mp-570835 | 6 6 6 (M) | 45 |
| SiRh | mp-2287 | 6 6 5 (G) | 52 |
| $BaSn_5$ | mp-571353 | 5 5 4 (G) | 50 |
| PS | mp-612 | 4 4 3 (G) | 64 |
| AsRh | mp-22079 | 8 5 4 (G) | 52 |
| $Te_2Au$ | mp-567525 | 7 7 5 (G) | 17 |
| $Be_3N_2$ | mp-6977 | 10 10 3 (G) | 46 |
| $Y_2O_3$ | mp-558573 | 8 4 3 (G) | 96 |
| AgO | mp-1079720 | 8 5 5 (G) | 52 |
| $FeTe_2$ | mp-1102002 | 4 4 4 (M) | 68 |
| $H_2O$ | mp-2739191 | 7 7 4 (G) | 24 |
| $CdO_2$ | mp-2310 | 5 5 5 (G) | 68 |
| $Mo_3Se_4$ | mp-21021 | 4 4 4 (M) | 86 |
| PbSe | mp-22009 | 6 6 2 (G) | 26 |
| $Be_{12}Pd$ | mp-12666 | 6 6 6 (M) | 69 |
| $Sr_2As$ | mp-15697 | 6 6 3 (G) | 28 |
| $AsRh_2$ | mp-1102364 | 7 4 3 (G) | 88 |
| ZrRh | mp-2808 | 8 8 8 (M) | 19 |
| $K_2S$ | mp-559278 | 5 5 4 (G) | 28 |
| $Ge_4Rh$ | mp-1104286 | 4 4 3 (G) | 135 |
| **Theoretical, metastable** ($n = 60$) | | | |
| $Li_3Ag$ | mp-976408 | 7 7 7 (G) | 24 |
| $Y_3Co$ | mp-1189329 | 3 4 4 (G) | 156 |
| TiNi | mp-1067248 | 10 7 6 (G) | 36 |
| TaP | mp-1187244 | 9 9 9 (G) | 13 |
| $Sr_3As_4$ | mp-678007 | 3 3 4 (G) | 62 |
| $Ga_2Se_3$ | mp-1224809 | 5 5 5 (G) | 30 |
| MnN | mp-1009130 | 10 10 10 (G) | 13 |
| $BeCl_2$ | mp-1190080 | 3 3 3 (G) | 78 |
| MgSn | mp-1094801 | 8 8 6 (M) | 13 |
| TiW | mp-1216621 | 10 10 6 (M) | 18 |
| $SrTl_3$ | mp-1206636 | 5 5 5 (G) | 32 |
| $IrCl_3$ | mp-729507 | 3 3 6 (G) | 84 |
| $Ta_4C_3$ | mp-1218000 | 6 6 6 (M) | 48 |
| $Sc_2O_3$ | mp-755313 | 6 5 6 (G) | 64 |
| $SnBr_2$ | mp-978985 | 4 4 6 (M) | 34 |
| $Sn_7Pd_5$ | mp-1219006 | 5 5 3 (G) | 108 |
| $Tl_4Sn$ | mp-1216560 | 5 5 5 (G) | 45 |
| $LiTl_3$ | mp-1185253 | 6 6 6 (M) | 32 |

Table S7 (continued)

| Formula | mp-id | k-point grid | NBANDS |
|---|---|---|---|
| $PtCl_2$ | mp-1219748 | 5 5 4 (G) | 34 |
| HCl | mp-684609 | 7 7 7 (G) | 5 |
| MgSn | mp-1094669 | 8 8 8 (M) | 13 |
| $K_5As_4$ | mp-684799 | 4 4 3 (G) | 41 |
| $KSn_3$ | mp-1185151 | 4 4 5 (G) | 64 |
| $NbS_2$ | mp-774873 | 10 6 2 (M) | 34 |
| $YAg_3$ | mp-1187811 | 6 6 6 (G) | 37 |
| $B_3Ir_2$ | mp-1228647 | 9 5 4 (G) | 60 |
| $TiMo_2$ | mp-1216675 | 5 6 14 (G) | 27 |
| $Cr_2C$ | mp-1226378 | 10 10 7 (G) | 22 |
| $ZrO_2$ | mp-775909 | 10 6 4 (M) | 36 |
| Zn | mp-2647117 | 11 11 11 (G) | 9 |
| $IN_2$ | mp-1097011 | 5 5 5 (G) | 24 |
| $CoSe_2$ | mp-1390196 | 4 4 4 (G) | 68 |
| GeS | mp-12910 | 7 7 5 (G) | 26 |
| ZrTi | mp-1215236 | 9 9 6 (G) | 19 |
| $VCo_3$ | mp-1095057 | 7 6 6 (G) | 72 |
| $Tl_3Ga$ | mp-1187539 | 5 5 5 (G) | 36 |
| $CaGe_2$ | mp-643542 | 7 7 6 (G) | 23 |
| $NbRh_2$ | mp-1220414 | 10 6 4 (M) | 54 |
| TcW | mp-1217337 | 10 10 6 (M) | 18 |
| $MgF_2$ | mp-560236 | 6 6 5 (G) | 24 |
| VN | mp-1017532 | 11 11 5 (G) | 26 |
| $Na_2Cl$ | mp-990084 | 6 6 6 (M) | 12 |
| $ZnPb_3$ | mp-1187949 | 5 5 5 (G) | 36 |
| $Sr_3Sn$ | mp-1187136 | 5 5 5 (G) | 24 |
| TaRe | mp-1217894 | 10 10 6 (M) | 18 |
| $Al_4Si$ | mp-1228120 | 6 6 6 (M) | 20 |
| $B_6Pb$ | mp-1096825 | 6 6 6 (M) | 33 |
| $ScTc_3$ | mp-867262 | 7 7 7 (G) | 37 |
| CrMo | mp-1226200 | 11 11 7 (G) | 18 |
| $BPd_5$ | mp-1228665 | 6 6 6 (M) | 49 |
| $Zr_3Be$ | mp-1188016 | 6 6 6 (M) | 35 |
| $Na_3Cl$ | mp-1064484 | 8 8 3 (G) | 16 |
| $SiAu_3$ | mp-1186998 | 7 7 7 (G) | 31 |
| $InSe_2$ | mp-1232036 | 7 7 3 (G) | 34 |
| $TiSi_2$ | mp-570991 | 8 8 8 (M) | 17 |
| $BeAu_3$ | mp-983539 | 7 7 7 (G) | 32 |
| $Zr_3Ge$ | mp-1188039 | 4 4 6 (G) | 78 |
| AlSb | mp-1023918 | 6 6 5 (G) | 16 |
| $Mg_{15}B$ | mp-1023515 | 4 4 3 (G) | 64 |
| HfMo | mp-1224314 | 10 10 6 (M) | 18 |

# E Elements and Allotropes

Every single-element structure with no matched polymorph in the dataset: the elemental references (`ELEMENT_REFERENCE`) used to decompose multi-element compounds. An experimental compound's formula is marked with a superscript star (black if it is the stable phase, red if it is a metastable polymorph/allotrope); unmarked formulas are theoretical. Auto-generated by `report/gen_appendix_extension.py`.

Table S8: Elements and allotropes with no matched polymorph in the `extension` campaign: references used to decompose multi-element compounds. Allotropes with one or more other allotropes computed in this dataset (e.g. carbon) are grouped in Table ?? instead. Black * = stable experimental reference (thermodynamic phase); red * = metastable experimental reference; unmarked = theoretical. No $\Delta$ICOHP column: an element is not itself a decomposition reaction.

| Formula | mp-id / COD | Space group | $E_{hull}$ (eV/at) |
|---|---|---|---|
| Ag* | mp-124 | Fm-3m | 0.0021 |
| Al* | mp-134 | Fm-3m | 0.0000 |
| As* | mp-158 | Cmce | 0.0000 |
| Au* | mp-81 | Fm-3m | 0.0000 |
| B* | mp-160 | R-3m | 0.0000 |
| Ba* | mp-122 | Im-3m | 0.0000 |
| Be* | mp-87 | P6_3/mmc | 0.0000 |
| Br* | mp-23154 | Cmce | 0.0000 |
| Ca* | mp-21 | Im-3m | 0.0056 |
| Cd* | mp-94 | P6_3/mmc | 0.0000 |
| $Cl_2$* | mp-22848 | Cmce | 0.0000 |
| Co* | mp-102 | Fm-3m | 0.0000 |
| Cr* | mp-90 | Im-3m | 0.0000 |
| Cs* | mp-1 | Im-3m | 0.0193 |
| Cu* | mp-30 | Fm-3m | 0.0000 |
| $F_2$* | mp-561203 | C2/c | 0.0031 |
| Fe* | mp-13 | Im-3m | 0.0000 |
| Ga* | mp-142 | Cmce | 0.0000 |
| Ge* | mp-32 | Fd-3m | 0.0000 |
| $H_2$* | mp-730101 | P2_12_12_1 | 0.0000 |
| Hf* | mp-103 | P6_3/mmc | 0.0000 |
| Hg* | mp-10861 | P6/mmm | 0.0030 |
| I* | mp-23153 | Cmce | 0.0000 |
| In* | mp-1055994 | I4/mmm | 0.0045 |
| Ir* | mp-101 | Fm-3m | 0.0000 |
| Li* | mp-1018134 | R-3m | 0.0000 |
| Mg* | mp-153 | P6_3/mmc | 0.0000 |
| Mn* | mp-35 | I-43m | 0.0000 |
| Mo* | mp-129 | Im-3m | 0.0000 |
| $N_2$ | mp-1059834 | Imma | 1.1010 |
| Nb* | mp-75 | Im-3m | 0.0000 |
| Ni* | mp-23 | Fm-3m | 0.0000 |
| $O_2$* | mp-1524462 | C2/m | 0.0000 |
| Os* | mp-49 | P6_3/mmc | 0.0000 |
| P* | mp-568348 | P2/c | 0.0000 |
| Pb* | mp-20483 | Fm-3m | 0.0000 |
| Pd* | mp-2 | Fm-3m | 0.0000 |
| Pt* | mp-126 | Fm-3m | 0.0000 |
| Rb* | mp-70 | Im-3m | 0.0092 |
| Re* | mp-8 | P6_3/mmc | 0.0036 |
| Rh* | mp-74 | Fm-3m | 0.0000 |
| Ru* | mp-33 | P6_3/mmc | 0.0000 |
| S* | mp-77 | Fddd | 0.0000 |
| Sb* | mp-104 | R-3m | 0.0000 |
| Sc* | mp-67 | P6_3/mmc | 0.0000 |
| Se* | mp-14 | P3_121 | 0.0011 |
| Sn* | mp-623511 | I4/mmm | 0.0614 |

| Formula | mp-id / COD | Space group | $E_{hull}$ (eV/at) |
|---|---|---|---|
| Sr* | mp-139 | P6_3/mmc | 0.0000 |
| Ta* | mp-50 | Im-3m | 0.0091 |
| Tc* | mp-113 | P6_3/mmc | 0.0000 |
| Te* | mp-19 | P3_121 | 0.0000 |
| Ti* | mp-46 | P6_3/mmc | 0.0152 |
| Tl* | mp-82 | P6_3/mmc | 0.0000 |
| V* | mp-146 | Im-3m | 0.0000 |
| W* | mp-91 | Im-3m | 0.0000 |
| Y* | mp-112 | P6_3/mmc | 0.0000 |
| Zn* | mp-79 | P6_3/mmc | 0.0000 |
| Zr* | mp-131 | P6_3/mmc | 0.0000 |

# F Polymorphs and Allotropes

Elements and compounds with several entries sharing a formula in the dataset, grouped here rather than scattered across the elements/compounds tables (see the table caption for detail).

Table S9: Elements (allotropes) and compounds (polymorphs) with **several entries sharing a formula** in the `extension` campaign (54 groups, 130 entries) — grouped here rather than scattered across the elements/compounds tables, since the natural comparison for a polymorph group is polymorph-to-polymorph, not the decomposition-to-elements reaction (hence no ΔICOHP column). A rule separates each group; within a group, sorted by increasing $E_{hull}$. Black * = stable experimental; red * = metastable experimental; unmarked = theoretical.

| Formula | mp-id / COD | Space group | $E_{hull}$ (eV/at) |
|---|---|---|---|
| AgN* | mp-1091417 | Cmce | 1.5201 |
| AgN | mp-1009766 | F-43m | 1.5495 |
| AgN | mp-1428312 | Pm-3m | 1.6519 |
| AgN | mp-1424734 | Ama2 | 1.9196 |
| $BaF_3$* | mp-1080821 | P2_1/m | 0.0000 |
| $BaF_3$ | mp-1183303 | I4/mmm | 0.0612 |
| BaO* | mp-1008500 | P6_3/mmc | 0.0189 |
| BaO | mp-1950989 | P6_3/mmc | 0.0434 |
| $Be_3N_2$* | mp-18337 | Ia-3 | 0.0000 |
| $Be_3N_2$ | mp-1017550 | P-3m1 | 0.2081 |
| $Be_3P_2$* | mp-567841 | Ia-3 | 0.0000 |
| $Be_3P_2$ | mp-1084771 | Pn-3m | 0.1567 |
| $BeCl_2$* | mp-23267 | Ibam | 0.0087 |
| $BeCl_2$ | mp-1172909 | I-43m | 3.1688 |
| $BeF_2$* | mp-15951 | P3_121 | 0.0004 |
| $BeF_2$ | mp-975644 | P6/mmm | 2.5181 |
| BeO | mp-1778 | F-43m | 0.0070 |
| BeO* | mp-7599 | P4_2/mnm | 0.0153 |
| $Bi_2O_3$* | mp-558953 | P2_1/c | 0.6000 |
| $Bi_2O_3$ | mp-674644 | P1 | 0.6199 |
| $Bi_2O_3$ | mp-1383505 | P1 | 0.9068 |
| $Bi_2O_3$ | mp-714859 | P1 | 2.4029 |

| Formula | mp-id / COD | Space group | $E_{hull}$ (eV/at) |
|---|---|---|---|
| C* | mp-48 | P6_3/mmc | 0.0031 |
| C* | mp-66 | Fd-3m | 0.1123 |
| C* | mp-47 | P6_3/mmc | 0.1395 |
| C | mp-1190171 | Pnma | 0.2927 |
| C | mp-1080826 | C2/m | 0.3009 |
| $C_3N_4$* | mp-571653 | P-43m | 0.4981 |
| $C_3N_4$ | mp-2852 | I-43d | 0.5033 |
| $C_3N_4$ | mp-1182484 | P6_3/m | 0.5848 |
| $C_3N_4$ | mp-563 | Fd-3m | 1.5119 |
| $Ca_3N_2$* | mp-844 | Ia-3 | 0.0000 |
| $Ca_3N_2$* | mp-1047 | R-3c | 0.0167 |
| $Ca_3N_2$ | mp-1013524 | Pm-3m | 0.5112 |
| $CaCl_2$* | mp-23214 | Pnnm | 0.0011 |
| $CaCl_2$ | mp-1120739 | Fm-3m | 0.0422 |
| $CaF_2$* | mp-10464 | Pnma | 0.0346 |
| $CaF_2$ | mp-554355 | P4/mmm | 0.2403 |
| CaO* | mp-2605 | Fm-3m | 0.0000 |
| CaO* | mp-1079707 | Cmce | 0.2098 |
| CaO | mp-1181903 | Fd-3m | 2.7281 |
| $CsN_3$* | mp-510557 | I4/mcm | 0.0000 |
| $CsN_3$ | mp-1225892 | P4/mmm | 0.0113 |
| $CsO_2$* | mp-1441 | I4/mmm | 0.0349 |
| $CsO_2$ | mp-1096936 | Cmmm | 0.0940 |
| $CsP_7$* | mp-1201790 | Pbcm | 0.0000 |
| $CsP_7$ | mp-1213206 | Pca2_1 | 0.0001 |
| $K_2O$* | mp-971 | Fm-3m | 0.0000 |
| $K_2O$ | mp-1223784 | P4/mmm | 0.2000 |
| $KF_3$ | mp-2749823 | P2_1 | 0.0119 |
| $KF_3$* | mp-1544557 | P2_12_12_1 | 0.0125 |
| $KN_2$* | mp-1071868 | Cmc2_1 | 1.1630 |
| $KN_2$ | mp-1180757 | Cmcm | 1.2293 |
| KP* | mp-1058315 | R-3m | 1.0080 |
| KP | mp-1057787 | Immm | 1.0092 |
| $Li_2O$* | mp-1960 | Fm-3m | 0.0000 |
| $Li_2O$ | mp-755894 | Pnma | 0.0844 |
| $Li_2S$* | mp-1125 | Pnma | 0.0615 |
| $Li_2S$ | mp-557142 | P6_3/mmc | 0.1750 |
| LiCl | mp-1185319 | P6_3mc | 0.0039 |
| LiCl* | mp-22905 | Fm-3m | 0.0243 |
| LiF* | mp-1138 | Fm-3m | 0.0000 |
| LiF | mp-1009009 | Pm-3m | 0.2875 |
| $LiP_5$* | mp-2412 | Pna2_1 | 0.0077 |
| $LiP_5$ | mp-32760 | Pna2_1 | 0.0963 |
| $MgCl_2$* | mp-23210 | R-3m | 0.0000 |
| $MgCl_2$ | mp-569626 | Ama2 | 0.1991 |
| $MgF_2$* | mp-1072956 | Pnnm | 0.0025 |
| $MgF_2$ | mp-1062842 | Amm2 | 0.2636 |

| Formula | mp-id / COD | Space group | $E_{hull}$ (eV/at) |
|---|---|---|---|
| MgO* | mp-549706 | P6_3mc | 0.1180 |
| MgO | mp-1009127 | Pm-3m | 0.7484 |
| MgS* | mp-1315 | Fm-3m | 0.0000 |
| MgS | mp-13032 | F-43m | 0.0343 |
| $Na_2O$* | mp-2352 | Fm-3m | 0.0000 |
| $Na_2O$ | mp-1975297 | C2/m | 0.0981 |
| $NaN_3$* | mp-22003 | R-3m | 0.0017 |
| $NaN_3$ | mp-1065265 | C2/m | 1.0162 |
| NaS* | mp-409 | P-62m | 0.0023 |
| NaS | mp-1180106 | C2/m | 1.0061 |
| NbO* | mp-2692 | Fm-3m | 0.7888 |
| NbO | mp-634965 | P4/mmm | 0.8240 |
| NbP* | mp-9830 | I4_1/amd | 0.6021 |
| NbP | mp-1220317 | Cmmm | 0.6209 |
| OsC* | mp-7142 | P-6m2 | 0.9161 |
| OsC | mp-999308 | P6_3/mmc | 0.9278 |
| OsC | mp-1009539 | Fm-3m | 1.3918 |
| OsC | mp-1009537 | Pm-3m | 1.4795 |
| $Rb_2O$* | mp-1394 | Fm-3m | 0.0032 |
| $Rb_2O$ | mp-2023482 | P-3m1 | 0.0479 |
| RbF | mp-11718 | Fm-3m | 0.0000 |
| RbF* | mp-2064 | Pm-3m | 0.0516 |
| $RbN_3$* | mp-581833 | P4/mmm | 0.0193 |
| $RbN_3$ | mp-1219567 | Amm2 | 0.7966 |
| RbP* | mp-1179688 | C2/m | 0.9879 |
| RbP | mp-1058499 | Immm | 0.9882 |
| RbS* | mp-558071 | Immm | 0.0019 |
| RbS | mp-1179684 | C2/m | 0.9975 |
| ReC* | mp-1079494 | P6_3/mmc | 0.6362 |
| ReC | mp-1009731 | Pm-3m | 1.1150 |
| RuC* | mp-10030 | P-6m2 | 0.6494 |
| RuC | mp-1009787 | Fm-3m | 0.9097 |
| SN* | COD:7017102 | P2_1/c | – |
| SN* | mp-236 | P2_1/c | 0.5087 |
| SN | mp-684642 | P2_1 | 0.5498 |
| SN | mp-1173453 | P2_1 | 0.7225 |
| SN | mp-1078816 | P2_1/c | 0.8090 |
| $SiO_2$* | mp-9258 | Pa-3 | 0.5770 |
| $SiO_2$ | mp-10064 | Fm-3m | 1.2779 |
| $SiO_2$ | mp-556044 | P-31c | 1.4574 |
| $SiO_2$ | mp-638049 | F4_132 | 1.7448 |
| $Sr_3P_4$* | mp-14288 | Fdd2 | 0.0000 |
| $Sr_3P_4$ | mp-682953 | P1 | 0.0081 |
| $SrF_2$ | mp-1102911 | Pnma | 0.0267 |
| $SrF_2$* | mp-1102240 | Pnma | 0.0646 |
| $SrO_2$* | mp-726705 | Pnma | 0.0096 |
| $SrO_2$ | mp-2763179 | Pnma | 0.0110 |

| Formula | mp-id / COD | Space group | $E_{hull}$ (eV/at) |
|---|---|---|---|
| TaN* | mp-1497 | P6/mmm | 0.4516 |
| TaN | mp-1009833 | F-43m | 0.4917 |
| TaN | mp-1009831 | Pm-3m | 0.8163 |
| $TiO_2$* | mp-1439 | Pbcn | 0.0045 |
| $TiO_2$* | mp-430 | P2_1/c | 0.0395 |
| $TiO_2$* | mp-9173 | Pnma | 0.0575 |
| TlC* | mp-567465 | Fm-3m | 2.1080 |
| TlC | mp-1058887 | P4/mmm | 2.3757 |
| TlN* | mp-1017982 | P6_3mc | 0.7334 |
| TlN | mp-1007778 | F-43m | 0.7337 |
| TlN | mp-1002222 | Fm-3m | 0.9712 |
| WC* | mp-13136 | Fm-3m | 0.4477 |
| WC | mp-1008635 | F-43m | 0.6729 |
| WC | mp-1008630 | Pm-3m | 0.8990 |

## G Target Compounds

Every multi-element target compound with no matched polymorph in the dataset, with ΔICOHP per atom and its endobondic/exobondic label where a case-1 decomposition reaction could be computed. Auto-generated from `analysis/reaction_icohp_case1.csv` and `analysis/reaction_analysis_ca` by `report/gen_appendix_extension.py`.

Table S10: Target compounds with no matched polymorph in the `extension` campaign. Formulas with several entries in this dataset are grouped in Table ?? instead. Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ΔICOHP per atom, products − reactants convention (`reaction_analysis`); "–" where no case-1 reaction could be computed (missing elemental reference).

| Formula | mp-id / COD | Space group | $E_{hull}$ (eV/at) | ΔICOHP/at. (eV) | Label |
|---|---|---|---|---|---|
| $Ba_5P_4$* | mp-17566 | Pnma | 0.0000 | -0.9590 | exobondic |
| $BaCl_2$* | mp-23199 | Pnma | 0.0068 | -0.2244 | exobondic |
| $BaN_2$* | mp-1102371 | Cc | 2.0568 | -8.6248 | exobondic |
| $BaS_3$* | mp-556296 | P2_12_12 | 0.0436 | -0.6233 | exobondic |
| CaN* | mp-1058549 | Fm-3m | 0.3982 | -4.9183 | exobondic |
| $Cs_2S$* | mp-1077814 | P6_3/mmc | 0.0667 | 0.9554 | endobondic |
| CsCl* | mp-573697 | Fm-3m | 0.0071 | 0.8830 | endobondic |
| CsF* | mp-8455 | Pm-3m | 0.1088 | 3.8464 | endobondic |
| $K_2S$* | mp-1022 | Fm-3m | 0.0000 | -0.3359 | exobondic |
| KCl* | mp-23289 | Pm-3m | 0.0776 | 0.7616 | endobondic |
| $KN_3$ | mp-636056 | I4/mcm | 1.0370 | -5.9991 | exobondic |
| $Li_3N$* | mp-2341 | P6_3/mmc | 0.0039 | 0.6165 | endobondic |
| $Mn_2O_7$* | mp-28338 | P2_1/c | 0.3346 | -1.4687 | exobondic |
| $MnO_2$* | mp-510408 | P4_2/mnm | 0.0464 | -0.5274 | exobondic |
| $Na_2Cl$ | mp-1077015 | Cmmm | 0.2391 | -0.3965 | exobondic |
| $PbN_6$* | mp-667338 | Pnma | 0.1093 | -2.5257 | exobondic |
| RbCl* | mp-23299 | Pm-3m | 0.0298 | 1.7924 | endobondic |
| $S_2N$* | COD:4031496 | P4_2nm | – | -0.9767 | exobondic |
| SrN* | mp-1058586 | Fm-3m | 0.4343 | -4.5395 | exobondic |
| SrS* | mp-10627 | Pm-3m | 0.2996 | -0.1088 | exobondic |

# H Endobondic Reactions

The subset of case-1 decomposition reactions ($A_aB_b \to a\,A + b\,B$, standard-state elemental references, balanced by `reaction_icohp.py`) labeled **endobondic** ($\Delta$ICOHP $\geq 0$, products − reactants convention — note this is the sign *opposite* `reaction_icohp.py`'s own `delta_icohp_per_atom` column reported elsewhere in this project; see `delta.py`'s docstring).

Table S11: Decomposition-into-elements (case 1) reactions labeled **endobondic** ($\Delta$ICOHP $\geq 0$, products − reactants) among the 208 compounds of the `extension` campaign. * = experimental starting compound (unmarked = theoretical).

| Reaction (balanced) | mp-id / COD | $E_{hull}$ (eV/at) | $\Delta$ICOHP/at. (eV) |
|---|---|---|---|
| 0.3333 $Be_3N_2 \to Be + 0.3333\ N_2$ | mp-1017550 | 0.2081 | 0.3960 |
| $BeCl_2{}^* \to Be + Cl_2$ | mp-23267 | 0.0087 | 1.8204 |
| $BeF_2{}^* \to Be + F_2$ | mp-15951 | 0.0004 | 3.1514 |
| $BeO^* \to Be + 0.5\ O_2$ | mp-7599 | 0.0153 | 2.7864 |
| $BeO \to Be + 0.5\ O_2$ | mp-1778 | 0.0070 | 4.4391 |
| 0.3333 $C_3N_4{}^* \to C + 0.6667\ N_2$ | mp-571653 | 0.4981 | 1.1208 |
| 0.3333 $C_3N_4 \to C + 0.6667\ N_2$ | mp-2852 | 0.5033 | 1.8456 |
| 0.5 $Cs_2S^* \to Cs + 0.5\ S$ | mp-1077814 | 0.0667 | 0.9554 |
| $CsCl^* \to Cs + 0.5\ Cl_2$ | mp-573697 | 0.0071 | 0.8830 |
| $CsF^* \to Cs + 0.5\ F_2$ | mp-8455 | 0.1088 | 3.8464 |
| $CsO_2{}^* \to Cs + O_2$ | mp-1441 | 0.0349 | 0.0666 |
| $CsO_2 \to Cs + O_2$ | mp-1096936 | 0.0940 | 0.7743 |
| $KCl^* \to K + 0.5\ Cl_2$ | mp-23289 | 0.0776 | 0.7616 |
| 0.5 $Li_2O^* \to Li + 0.25\ O_2$ | mp-1960 | 0.0000 | 2.3366 |
| 0.5 $Li_2O \to Li + 0.25\ O_2$ | mp-755894 | 0.0844 | 1.9481 |
| 0.5 $Li_2S^* \to Li + 0.5\ S$ | mp-1125 | 0.0615 | 1.8505 |
| 0.5 $Li_2S \to Li + 0.5\ S$ | mp-557142 | 0.1750 | 1.5052 |
| 0.3333 $Li_3N^* \to Li + 0.1667\ N_2$ | mp-2341 | 0.0039 | 0.6165 |
| $LiCl^* \to Li + 0.5\ Cl_2$ | mp-22905 | 0.0243 | 2.8030 |
| $LiCl \to Li + 0.5\ Cl_2$ | mp-1185319 | 0.0039 | 2.5005 |
| $LiF^* \to Li + 0.5\ F_2$ | mp-1138 | 0.0000 | 4.4673 |
| $LiF \to Li + 0.5\ F_2$ | mp-1009009 | 0.2875 | 4.5131 |
| $LiP_5{}^* \to Li + 5\ P$ | mp-2412 | 0.0077 | 0.9043 |
| $LiP_5 \to Li + 5\ P$ | mp-32760 | 0.0963 | 0.8692 |
| $MgCl_2{}^* \to Mg + Cl_2$ | mp-23210 | 0.0000 | 0.1568 |
| $MgF_2{}^* \to Mg + F_2$ | mp-1072956 | 0.0025 | 0.0015 |
| $MgF_2 \to Mg + F_2$ | mp-1062842 | 0.2636 | 0.0229 |
| 0.5 $Rb_2O^* \to Rb + 0.25\ O_2$ | mp-1394 | 0.0032 | 0.7235 |
| $RbCl^* \to Rb + 0.5\ Cl_2$ | mp-23299 | 0.0298 | 1.7924 |
| $RbF^* \to Rb + 0.5\ F_2$ | mp-2064 | 0.0516 | 3.9244 |
| $RbF \to Rb + 0.5\ F_2$ | mp-11718 | 0.0000 | 1.9497 |
| $SiO_2{}^* \to Si + O_2$ | mp-9258 | 0.5770 | 3.3332 |
| $SiO_2 \to Si + O_2$ | mp-10064 | 1.2779 | 2.1735 |
| $SiO_2 \to Si + O_2$ | mp-556044 | 1.4574 | 0.0958 |
| $SiO_2 \to Si + O_2$ | mp-638049 | 1.7448 | 0.3849 |

# I Exobondic Reactions

The complementary **exobondic** subset ($\Delta$ICOHP $< 0$), same convention and provenance as Appendix **??**.

Table S12: Decomposition-into-elements (case 1) reactions labeled **exobondic** ($\Delta$ICOHP $< 0$, products − reactants) among the 208 compounds of the `extension` campaign. * = experimental starting compound (unmarked = theoretical).

| Reaction (balanced) | mp-id / COD | $E_{hull}$ (eV/at) | ΔICOHP/at. (eV) |
|---|---|---|---|
| $AgN^* \rightarrow Ag + 0.5\ N_2$ | mp-1091417 | 1.5201 | -7.5875 |
| $AgN \rightarrow Ag + 0.5\ N_2$ | mp-1009766 | 1.5495 | -6.4957 |
| $AgN \rightarrow Ag + 0.5\ N_2$ | mp-1424734 | 1.9196 | -8.0907 |
| $AgN \rightarrow Ag + 0.5\ N_2$ | mp-1428312 | 1.6519 | -5.3559 |
| $0.2\ Ba_5P_4{}^* \rightarrow Ba + 0.8\ P$ | mp-17566 | 0.0000 | -0.9590 |
| $BaCl_2{}^* \rightarrow Ba + Cl_2$ | mp-23199 | 0.0068 | -0.2244 |
| $BaF_3{}^* \rightarrow Ba + 1.5\ F_2$ | mp-1080821 | 0.0000 | -0.8680 |
| $BaF_3 \rightarrow Ba + 1.5\ F_2$ | mp-1183303 | 0.0612 | -0.2493 |
| $BaN_2{}^* \rightarrow Ba + N_2$ | mp-1102371 | 2.0568 | -8.6248 |
| $BaO^* \rightarrow Ba + 0.5\ O_2$ | mp-1008500 | 0.0189 | -0.3221 |
| $BaO \rightarrow Ba + 0.5\ O_2$ | mp-1950989 | 0.0434 | -1.6129 |
| $BaS_3{}^* \rightarrow Ba + 3\ S$ | mp-556296 | 0.0436 | -0.6233 |
| $0.3333\ Be_3N_2{}^* \rightarrow Be + 0.3333\ N_2$ | mp-18337 | 0.0000 | -0.0746 |
| $0.3333\ Be_3P_2{}^* \rightarrow Be + 0.6667\ P$ | mp-567841 | 0.0000 | -0.2111 |
| $0.3333\ Be_3P_2 \rightarrow Be + 0.6667\ P$ | mp-1084771 | 0.1567 | -0.1795 |
| $BeCl_2 \rightarrow Be + Cl_2$ | mp-1172909 | 3.1688 | -5.7807 |
| $BeF_2 \rightarrow Be + F_2$ | mp-975644 | 2.5181 | -0.6816 |
| $0.3333\ C_3N_4 \rightarrow C + 0.6667\ N_2$ | mp-1182484 | 0.5848 | -3.6399 |
| $0.3333\ C_3N_4 \rightarrow C + 0.6667\ N_2$ | mp-563 | 1.5119 | -0.4498 |
| $0.3333\ Ca_3N_2{}^* \rightarrow Ca + 0.3333\ N_2$ | mp-1047 | 0.0167 | -4.4881 |
| $0.3333\ Ca_3N_2{}^* \rightarrow Ca + 0.3333\ N_2$ | mp-844 | 0.0000 | -4.4374 |
| $0.3333\ Ca_3N_2 \rightarrow Ca + 0.3333\ N_2$ | mp-1013524 | 0.5112 | -4.4898 |
| $CaCl_2{}^* \rightarrow Ca + Cl_2$ | mp-23214 | 0.0011 | -0.3199 |
| $CaCl_2 \rightarrow Ca + Cl_2$ | mp-1120739 | 0.0422 | -0.2208 |
| $CaF_2{}^* \rightarrow Ca + F_2$ | mp-10464 | 0.0346 | -0.2227 |
| $CaF_2 \rightarrow Ca + F_2$ | mp-554355 | 0.2403 | -0.1814 |
| $CaN^* \rightarrow Ca + 0.5\ N_2$ | mp-1058549 | 0.3982 | -4.9183 |
| $CaO^* \rightarrow Ca + 0.5\ O_2$ | mp-1079707 | 0.2098 | -2.2127 |
| $CaO^* \rightarrow Ca + 0.5\ O_2$ | mp-2605 | 0.0000 | -0.9525 |
| $CaO \rightarrow Ca + 0.5\ O_2$ | mp-1181903 | 2.7281 | -2.9971 |
| $CsN_3{}^* \rightarrow Cs + 1.5\ N_2$ | mp-510557 | 0.0000 | -2.2903 |
| $CsN_3 \rightarrow Cs + 1.5\ N_2$ | mp-1225892 | 0.0113 | -1.6664 |
| $CsP_7{}^* \rightarrow Cs + 7\ P$ | mp-1201790 | 0.0000 | -0.2978 |
| $CsP_7 \rightarrow Cs + 7\ P$ | mp-1213206 | 0.0001 | -0.2667 |
| $0.5\ K_2O^* \rightarrow K + 0.25\ O_2$ | mp-971 | 0.0000 | -0.4357 |
| $0.5\ K_2O \rightarrow K + 0.25\ O_2$ | mp-1223784 | 0.2000 | -1.5638 |
| $0.5\ K_2S^* \rightarrow K + 0.5\ S$ | mp-1022 | 0.0000 | -0.3359 |
| $KF_3{}^* \rightarrow K + 1.5\ F_2$ | mp-1544557 | 0.0125 | -1.1511 |
| $KF_3 \rightarrow K + 1.5\ F_2$ | mp-2749823 | 0.0119 | -1.1505 |
| $KN_2{}^* \rightarrow K + N_2$ | mp-1071868 | 1.1630 | -3.4244 |
| $KN_2 \rightarrow K + N_2$ | mp-1180757 | 1.2293 | -4.2865 |
| $KN_3 \rightarrow K + 1.5\ N_2$ | mp-636056 | 1.0370 | -5.9991 |
| $KP^* \rightarrow K + P$ | mp-1058315 | 1.0080 | -2.1600 |
| $KP \rightarrow K + P$ | mp-1057787 | 1.0092 | -2.1366 |
| $MgCl_2 \rightarrow Mg + Cl_2$ | mp-569626 | 0.1991 | -0.0862 |
| $MgO^* \rightarrow Mg + 0.5\ O_2$ | mp-549706 | 0.1180 | -1.4179 |
| $MgO \rightarrow Mg + 0.5\ O_2$ | mp-1009127 | 0.7484 | -1.0763 |
| $MgS^* \rightarrow Mg + S$ | mp-1315 | 0.0000 | -0.3401 |
| $MgS \rightarrow Mg + S$ | mp-13032 | 0.0343 | -0.4163 |
| $0.5\ Mn_2O_7{}^* \rightarrow Mn + 1.75\ O_2$ | mp-28338 | 0.3346 | -1.4687 |
| $MnO_2{}^* \rightarrow Mn + O_2$ | mp-510408 | 0.0464 | -0.5274 |

| Reaction (balanced) | mp-id / COD | $E_{hull}$ (eV/at) | ΔICOHP/at. (eV) |
|---|---|---|---|
| 0.5 $Na_2Cl \rightarrow Na + 0.25\ Cl_2$ | mp-1077015 | 0.2391 | -0.3965 |
| 0.5 $Na_2O^* \rightarrow Na + 0.25\ O_2$ | mp-2352 | 0.0000 | -1.3057 |
| 0.5 $Na_2O \rightarrow Na + 0.25\ O_2$ | mp-1975297 | 0.0981 | -1.5619 |
| $NaN_3{}^* \rightarrow Na + 1.5\ N_2$ | mp-22003 | 0.0017 | -1.5201 |
| $NaN_3 \rightarrow Na + 1.5\ N_2$ | mp-1065265 | 1.0162 | -5.4248 |
| $NaS^* \rightarrow Na + S$ | mp-409 | 0.0023 | -0.4911 |
| $NaS \rightarrow Na + S$ | mp-1180106 | 1.0061 | -2.3999 |
| $NbO^* \rightarrow Nb + 0.5\ O_2$ | mp-2692 | 0.7888 | -2.3179 |
| $NbO \rightarrow Nb + 0.5\ O_2$ | mp-634965 | 0.8240 | -3.1495 |
| $NbP^* \rightarrow Nb + P$ | mp-9830 | 0.6021 | -2.4724 |
| $NbP \rightarrow Nb + P$ | mp-1220317 | 0.6209 | -1.1731 |
| $OsC^* \rightarrow Os + C$ | mp-7142 | 0.9161 | -2.0168 |
| $OsC \rightarrow Os + C$ | mp-1009537 | 1.4795 | -2.3184 |
| $OsC \rightarrow Os + C$ | mp-1009539 | 1.3918 | -3.1141 |
| $OsC \rightarrow Os + C$ | mp-999308 | 0.9278 | -2.9572 |
| $PbN_6{}^* \rightarrow Pb + 3\ N_2$ | mp-667338 | 0.1093 | -2.5257 |
| 0.5 $Rb_2O \rightarrow Rb + 0.25\ O_2$ | mp-2023482 | 0.0479 | -0.7592 |
| $RbN_3{}^* \rightarrow Rb + 1.5\ N_2$ | mp-581833 | 0.0193 | -1.3478 |
| $RbN_3 \rightarrow Rb + 1.5\ N_2$ | mp-1219567 | 0.7966 | -3.5636 |
| $RbP^* \rightarrow Rb + P$ | mp-1179688 | 0.9879 | -1.9315 |
| $RbP \rightarrow Rb + P$ | mp-1058499 | 0.9882 | -1.9104 |
| $RbS^* \rightarrow Rb + S$ | mp-558071 | 0.0019 | -0.3217 |
| $RbS \rightarrow Rb + S$ | mp-1179684 | 0.9975 | -2.7347 |
| $ReC^* \rightarrow Re + C$ | mp-1079494 | 0.6362 | -2.6625 |
| $ReC \rightarrow Re + C$ | mp-1009731 | 1.1150 | -1.9955 |
| $RuC^* \rightarrow Ru + C$ | mp-10030 | 0.6494 | -2.1301 |
| $RuC \rightarrow Ru + C$ | mp-1009787 | 0.9097 | -2.8605 |
| 0.5 $S_2N^* \rightarrow S + 0.25\ N_2$ | COD:4031496 | – | -0.9767 |
| $SN^* \rightarrow S + 0.5\ N_2$ | COD:7017102 | – | -1.3963 |
| $SN^* \rightarrow S + 0.5\ N_2$ | mp-236 | 0.5087 | -2.3179 |
| $SN \rightarrow S + 0.5\ N_2$ | mp-1078816 | 0.8090 | -3.7082 |
| $SN \rightarrow S + 0.5\ N_2$ | mp-1173453 | 0.7225 | -3.1673 |
| $SN \rightarrow S + 0.5\ N_2$ | mp-684642 | 0.5498 | -1.7362 |
| 0.3333 $Sr_3P_4{}^* \rightarrow Sr + 1.333\ P$ | mp-14288 | 0.0000 | -1.0472 |
| 0.3333 $Sr_3P_4 \rightarrow Sr + 1.333\ P$ | mp-682953 | 0.0081 | -1.0793 |
| $SrF_2{}^* \rightarrow Sr + F_2$ | mp-1102240 | 0.0646 | -0.5421 |
| $SrF_2 \rightarrow Sr + F_2$ | mp-1102911 | 0.0267 | -0.1481 |
| $SrN^* \rightarrow Sr + 0.5\ N_2$ | mp-1058586 | 0.4343 | -4.5395 |
| $SrO_2{}^* \rightarrow Sr + O_2$ | mp-726705 | 0.0096 | -1.9205 |
| $SrO_2 \rightarrow Sr + O_2$ | mp-2763179 | 0.0110 | -1.7522 |
| $SrS^* \rightarrow Sr + S$ | mp-10627 | 0.2996 | -0.1088 |
| $TaN^* \rightarrow Ta + 0.5\ N_2$ | mp-1497 | 0.4516 | -5.9309 |
| $TaN \rightarrow Ta + 0.5\ N_2$ | mp-1009831 | 0.8163 | -4.0540 |
| $TaN \rightarrow Ta + 0.5\ N_2$ | mp-1009833 | 0.4917 | -5.3385 |
| $TiO_2{}^* \rightarrow Ti + O_2$ | mp-1439 | 0.0045 | -0.6461 |
| $TiO_2{}^* \rightarrow Ti + O_2$ | mp-9173 | 0.0575 | -0.6509 |
| $TiO_2{}^* \rightarrow Ti + O_2$ | mp-430 | 0.0395 | -0.6267 |
| $TlC^* \rightarrow Tl + C$ | mp-567465 | 2.1080 | -3.1404 |
| $TlC \rightarrow Tl + C$ | mp-1058887 | 2.3757 | -3.3638 |
| $TlN^* \rightarrow Tl + 0.5\ N_2$ | mp-1017982 | 0.7334 | -3.7498 |
| $TlN \rightarrow Tl + 0.5\ N_2$ | mp-1002222 | 0.9712 | -2.3228 |
| $TlN \rightarrow Tl + 0.5\ N_2$ | mp-1007778 | 0.7337 | -3.1948 |
| $WC^* \rightarrow W + C$ | mp-13136 | 0.4477 | -6.2684 |
| $WC \rightarrow W + C$ | mp-1008630 | 0.8990 | -5.9400 |
| $WC \rightarrow W + C$ | mp-1008635 | 0.6729 | -7.7315 |

# J ICOHP and ICOBI Antibonding Descriptors

## J.1 Implementation

The ICOHP antibonding-population-near-frontier descriptor (§??, `cohp_extraction.py`) is extended here to ICOBI: same one-sided window $(E_{\text{ref}} - \Delta E,\ E_{\text{ref}}]$, same $E_{\text{ref}}$ (Fermi level for metals, VBM for gapped compounds — logic entirely independent of which curve is integrated), read this time from `COBICAR.lobster`/`ICOBILIST.lobster` instead of `COHPCAR.lobster`/`ICOHPLIST.lobster` (`pymatgen.io.lobster.outputs.Cohpcar(are_cobis=True)`).

**Critical point, empirically verified, not assumed**: ICOBI does **not** share ICOHP's sign convention. Whereas "negative ICOHP = bonding" (§2.4), ICOBI follows the opposite convention, "positive ICOBI = bonding" (like COOP, not like COHP). Verified two ways on real data: (1) exact cross-check against `ICOBILIST.lobster` (max deviation 0.0 across 66 bond labels, `tests/test_cohp_extraction.py`) — `pymatgen` does not silently renormalize the sign when switching from COHP to COBI parsing; (2) on `extension_SiO2_anchor`'s strongest Si–O bond (ICOHP$= -5.599$, ICOBI$= +0.513$), the deep bonding region ($-10$ to $-3$ eV, far from any frontier effects) gives a mean COHP($E$) of $-0.338$ but a mean COBI($E$) of $+0.041$ — opposite signs in the same physically bonding-dominated region. ICOBI's antibonding part is therefore its *negative* part, not its positive part – the clip direction (`integrate_icobi_antibonding_in_window()`) is deliberately the mirror image of the ICOHP version, not a copy.

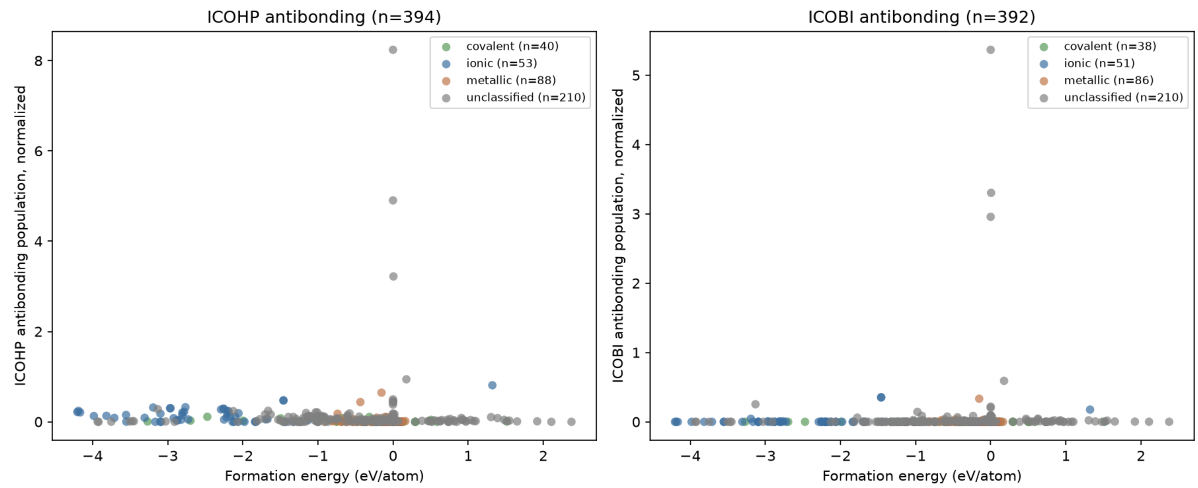
Computed across the 394 compounds with a `COBICAR.lobster` file (`analysis/compute_icobi_antibonding_` 392 with `is_metal` available.

## J.2 Correlation with formation energy

Table S13: Spearman correlations (normalized, $\Delta E = 1.0$ eV) against `formation_energy_per_atom` — ICOHP (n=394) and ICOBI (n=392) antibonding, same population.

| Group | $n$ (ICOHP) | $\rho$ (ICOHP) | $n$ (ICOBI) | $\rho$ (ICOBI) |
|---|---|---|---|---|
| all | 391 | $-0.2473^{***}$ | 385 | $+0.2518^{***}$ |
| `covalent` | 40 | $-0.4781^{**}$ | 38 | $-0.1755$ |
| `ionic` | 53 | $-0.1842$ | 51 | $+0.1289$ |
| `metallic` | 88 | $-0.1864$ | 86 | $+0.0011$ |
| `is_metal=False` | 166 | $-0.2621^{***}$ | 162 | $+0.0661$ |
| `is_metal=True` | 225 | $-0.0154$ | 223 | $+0.1263$ |

$^{**}p < 0.01$; $^{***}p < 0.001$; no star = not significant. Full detail (raw + normalized, all three $\Delta E$): `analysis/stats_summary_antibonding_joint.json`.




**The two descriptors behave differently.** ICOBI antibonding's global sign is the *opposite* of ICOHP antibonding's (+0.25 vs. −0.25) — for ICOHP, more occupied antibonding character near $E_F$ is associated with *more* stable formation (already noted as counterintuitive in §??); for ICOBI, the relationship runs the naively expected way (more antibonding ⇒ less stable). But **no ICOBI subgroup survives stratification** (covalent, ionic, metallic, `is_metal`: $p > 0.05$ throughout) — unlike ICOHP, whose covalent subgroup ($n = 40$, $\rho = -0.478$) remains the project's most robust stratified result. ICOBI's global correlation is therefore, like reaction ICOHP's (§??), likely a between-bond-type clustering effect rather than a robust relationship within a homogeneous group.

## J.3 Δ(antibonding) of the decomposition-into-elements reaction

For a case-1 reaction (compound → elements), Δ(antibonding) is **not** computed by `reaction_analysis/delta.p` (which combines *extensive* per-formula-unit ICOHP/ICOBI sums with integer coefficients): the antibonding population near the frontier is built from LOBSTER's own "average" trace, an *intensive* quantity (averaged over all bonds, not summed) with no natural extensive decomposition. Convention chosen here, documented as an explicit methodological choice rather than a silent reuse: Δ = (atomic-fraction-weighted average of the elements' own antibonding descriptors) − (the compound's own antibonding descriptor), still in products − reactants convention. Computed for 320 case-1 reactions (`analysis/compute_delta_antibonding_case1.py`).

Table S14: Spearman correlations, Δ(antibonding) of the decomposition reaction (case 1, $n =$ 316–320).

| Descriptor | Target | $n$ | $\rho$ |
|---|---|---|---|
| Δ(ICOHP antibonding) | `formation_energy_per_atom` | 316 | $-0.6439^{***}$ |
| Δ(ICOBI antibonding) | `formation_energy_per_atom` | 316 | $-0.4457^{***}$ |
| Δ(ICOHP antibonding) | `energy_above_hull` | 320 | $-0.1868^{***}$ |
| Δ(ICOBI antibonding) | `energy_above_hull` | 320 | $-0.1524^{**}$ |

$^{**}p < 0.01$; $^{***}p < 0.001$.

**Δ(ICOHP antibonding) is, at this stage, the strongest continuous correlation in the whole project** ($\rho = -0.644$, stronger than reaction ICOHP in §??, $\rho = 0.502$). Notable: unlike the raw values (opposite signs, above), **both Δs run the same direction** (both negative) — the delta construction appears to align the two descriptors on a consistent reading, where their raw values do not. The two Δs are themselves correlated with each other ($\rho = 0.532$, $p < 0.001$, $n = 320$): related but not redundant.

## J.4 Stratification by bond type

Table S15: Spearman correlations, $\Delta$(antibonding) against `formation_energy_per_atom`, by subgroup (case 1, $n = 316$).

| Subgroup | $n$ | $\rho$ ($\Delta$ICOHP) | $n$ | $\rho$ ($\Delta$ICOBI) |
|---|---|---|---|---|
| all | 316 | $-0.6439^{***}$ | 316 | $-0.4457^{***}$ |
| `covalent` | 29 | $-0.5798^{***}$ | 29 | $-0.2044$ |
| `ionic` | 51 | $-0.4205^{**}$ | 51 | $+0.0608$ |
| `metallic` | 86 | $-0.4858^{***}$ | 86 | $-0.4872^{***}$ |
| `is_metal=False` | 145 | $-0.6210^{***}$ | 145 | $-0.3248^{***}$ |
| `is_metal=True` | 171 | $-0.5490^{***}$ | 171 | $-0.4653^{***}$ |

$^{**}p < 0.01$; $^{***}p < 0.001$; no star = not significant.

**$\Delta$(ICOHP antibonding) is, to date, the first descriptor in the whole project whose global correlation survives *every* stratification tested** — covalent, ionic, metallic, `is_metal` on both sides, all significant at $p < 0.01$ or better. No other descriptor in this report manages that: the raw antibonding population (§**??**) keeps only its covalent subgroup, reaction ICOHP (§**??**) loses every `bond_type` stratum and survives only on `is_metal`. $\Delta$(ICOBI antibonding) is more mixed: survives on metallic and both `is_metal` sides but not on covalent ($p = 0.29$) or ionic ($p = 0.67$). This remains a newly chosen construction (previous subsection) — its magnitude will need to reproduce on an independent dataset before being treated as a fully established result, but surviving full stratification makes it, as it stands, the project's most robust result.

## J.5 Reaction $\Delta$(ICOHP) restricted to a window near $E_F$

Extension proposed in the Conclusion (§**??**): rather than ICOHP integrated over the whole occupied range (§**??**) or only the antibonding part near $E_F$ (previous subsections), a third construction — the *signed* ICOHP (bonding *and* antibonding, not clipped) but integrated *only* over the one-sided window $(E_F - \Delta E,\ E_F]$, $\Delta E = 1.0$ eV, summed over *every* bond label (extensive, like the full ICOHP sum) rather than just the "average" trace — then combined with the same reaction stoichiometry as the full $\Delta$(ICOHP) (`reaction_icohp.py`, reactant − predicted products convention). Implementation: `cohp_extraction.icohp_windowed_per_atom()` reads the difference between two points on the *cumulative* ICOHP curve LOBSTER already computes in `COHPCAR.lobster` (`ICOHP_cumulative`($E_F$) − `ICOHP_cumulative`($E_F - \Delta E$), per label), no new trapezoidal integration — validated: at very large $\Delta E$ (covering the whole calculation window), this quantity reproduces the full ICOHP sum to within $\sim$0.1%.

An intermediate result between the other two constructions: unlike the full reaction ICOHP (§**??**), which loses *every* `bond_type` stratum, this windowed version **survives on the ionic subgroup** ($\rho = 0.524$, $p < 0.001$) and on `is_metal` on both sides — but not on metallic, and only marginally on covalent ($p = 0.053$). Less robust than $\Delta$(ICOHP antibonding) (previous subsection, survives everywhere), more robust than the full reaction ICOHP. Cross-correlation with $\Delta$(ICOHP antibonding): $\rho = -0.599$ ($p < 0.001$, $n = 320$) — strongly *anti*correlated despite their closely related construction (same window, same $E_F$), a sign that clipping to the antibonding-only part (previous subsection) captures qualitatively different information from the full signed ICOHP in the same window. Cross-correlation with the full reaction ICOHP: $\rho = 0.231$ ($p < 0.001$) — related but clearly distinct, not simply a restriction of the same quantity.

Table S16: Spearman correlations, $\Delta$(windowed ICOHP near $E_F$) against `formation_energy_per_atom` (case 1, $n = 316$).

| Subgroup | $n$ | $\rho$ |
|---|---|---|
| all | 316 | $+0.3712^{***}$ |
| `covalent` | 29 | +0.3635 (p=0.053) |
| `ionic` | 51 | $+0.5243^{***}$ |
| `metallic` | 86 | +0.1721 |
| `is_metal=False` | 145 | $+0.4660^{***}$ |
| `is_metal=True` | 171 | $+0.1848^{*}$ |

$^{*}p < 0.05$; $^{***}p < 0.001$; no star = not significant. Against `energy_above_hull`, no significant correlation anywhere ($n = 320$, $p > 0.07$ throughout) — the same behavior as every other ICOHP/ICOBI descriptor in this report.

## K Reitz & Dronskowski reaction structures

The §?? validation (Table ??) only compares the arithmetic of the manuscript's own published ICOHP values — independent of any local crystal structure. The table below documents, for each of the 16 species involved in these 7 reactions, the structure as described by the manuscript (where it specifies one), the Materials Project/COD entry used in this dataset, and whether an independent verification was possible. Three cases revealed a real bug or ambiguity, since fixed: $N_2$ (a theoretical polymeric phase substituted for an isolated gaseous dimer, §??), $Pb(N_3)_2$ (the wrong Pnma polymorph substituted for the Pna2_1 the manuscript cites), and ZnSn's competing tin phase ($\beta$-tin substituted for $\alpha$-tin).

Table S17: The 16 species across Reitz & Dronskowski's 7 reactions (ic-2026-04181q, Table ??): structure as described by the manuscript, the MP/COD entry used in this dataset, and verification status.

| Species (role) | Manuscript structure | Our source | Verification |
|---|---|---|---|
| $Pb(N_3)_2$ (target) | lead azide, ICSD 28336/16887 (Glen 1963) | mp-620058, Pna2_1, $E_{hull} = 0.110$ | **Yes** — exact polymorph tified 2026-08-17, substi for mp-667338 (Pnma) al present in the dataset |
| Pb (element) | not specified (standard elemental reference) | mp-20483, Fm-3m, $E_{hull} = 0.000$, on-hull | not required — no polym ambiguity for fcc metallic I |
| $N_2$ (element, 4 reactions) | gaseous N≡N, $d \approx 1.10$ Å | `gasref_N2_dimerbox`, hand-built isolated dimer (10×10×10 Å box), $d$(N-N)= 1.113 Å, ICOHP= −22.998 eV | **Yes** — within 0.7% o manuscript's value (−23.16 Before 2026-08-17: mp-10 (theoretical, polymeric pha N= 1.296 Å) — a real bug, |
| $S_4N_2$ (target) | not specified beyond composition/ICOHP | COD 4031496, P4_2nm ($Z$ = 4), no MP entry exists for this composition | no — COD experimental ture used as-is, no ind dent confirmation it is manuscript's exact polymo |
| $S_8$ (element) | not specified | mp-77, Fddd ($\alpha$-$S_8$), $E_{hull} = 0.000$, on-hull | not required — unambi standard reference |
| $S_4N_4$ (target) | not specified | COD 7017102, P2_1/c ($Z = 4$) | no |

| Species (role) | Manuscript structure | Our source | Verification |
|---|---|---|---|
| ZnSn (target) | implicit: 6 Zn-Sn bonds + 1 Zn-Zn bond/formula unit (consistent with a NiAs prototype) | `gasref_ZnSn_NiAs`, hand-built, P6_3/mmc (#194), no MP/COD entry (no stable real Zn-Sn intermetallic exists) | partial — the (6+1) coordin is consistent with NiAs, b independent crystallograph erence was found to confir manuscript's exact lattic rameters |
| Zn (element) | not specified | mp-79, P6_3/mmc, $E_{hull} = 0.000$, on-hull | not required |
| Sn (element, ZnSn's competing phase) | implies a phase distinct from the dataset's "default" $\beta$-tin | mp-117, Fd-3m ($\alpha$-tin, diamond-cubic), substituted for mp-623511 ($\beta$-tin, I4/mmm) | partial — $\alpha$-tin (the temperature fcc form) is structurally consistent peting phase for ZnSn; independent confirmation manuscript uses mp-117 sp cally |
| CaO[sphalerite] (target) | F-43m (#216), explicitly cited | `gasref_CaO_sphalerite`, hand-built (no MP entry for this metastable CaO polymorph), F-43m confirmed by symmetry | **Yes** (space group) |
| CaO[rocksalt] (product) | Fm-3m (#225), explicitly cited | mp-2605, Fm-3m, $E_{hull} = 0.000$, on-hull | **Yes** |
| CaN (target) | not specified | mp-1058549, Fm-3m (rocksalt-type), $E_{hull} = 0.398$ | no |
| $Ca_3N_2$ (product) | not specified | mp-844, Ia-3 (anti-bixbyite), $E_{hull} = 0.000$, on-hull | no — the only on-hull entr most defensible default cho |
| $Mn_2O_7$ (target) | not specified | mp-28338, P2_1/c, $E_{hull} = 0.335$ | no |
| $MnO_2$ (product) | not specified | mp-510408, P4_2/mnm (rutile/pyrolusite) | no |
| $O_2$ (element) | gaseous $O_2$, triplet ground state | mp-1524462, C2/m, $E_{hull} = 0.000$, on-hull, dedicated spin-polarized treatment | structural — isolated dimer O)= 1.222 Å vs. 1.208 Å l ture (within 1%) |

**Reading key.** "Verification: Yes" means the structure was confirmed to match the manuscript's explicit description (a cited space group, or a recomputed bond distance compared directly). "Partial" means the geometry/chemistry is consistent with what the manuscript describes but no independent crystallographic citation allowed a strict confirmation. "No" means only the composition was verified: the manuscript does not specify a space group for that species, so the most reasonable MP/COD entry was used with no guarantee it is *the* exact structure the authors used. This does not invalidate the §?? validation (which only compares the arithmetic reconstruction of the manuscript's own published ICOHP values, independent of any local structure) but it does bound the scope of any future comparison between our own DFT+LOBSTER calculations and the manuscript's numbers for these species.

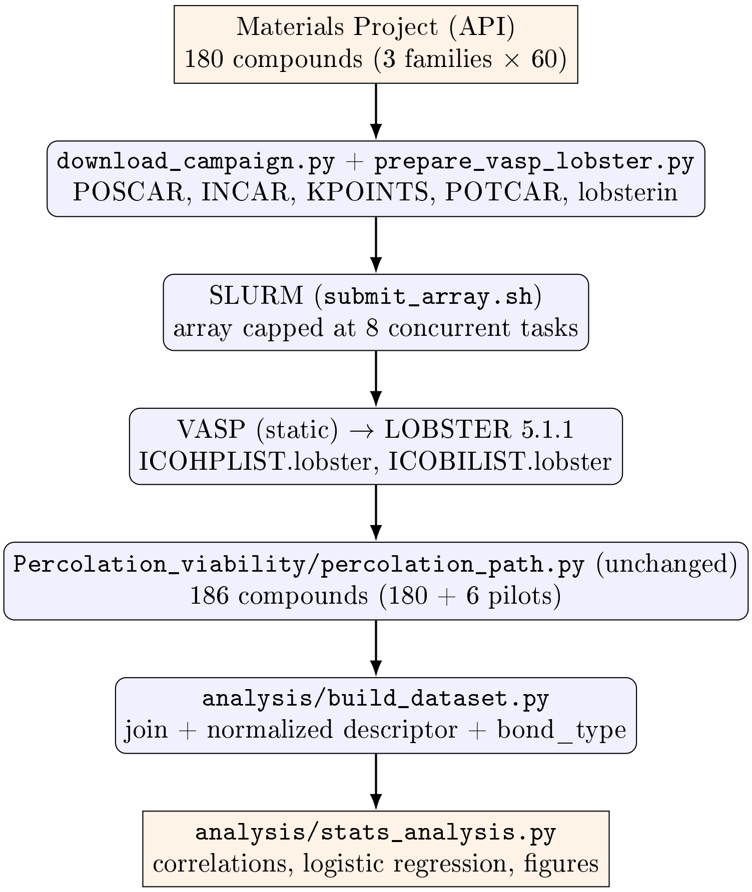
# L Percolation

This appendix collects everything related to the percolation weight and its direct descendants (network dimensionality, periodic min-cut — missions #1 and #3): this was the central subject of the original campaign 2, unmodified below from its first drafting, but it is no longer this project's organizing question (§??). The corresponding source code has been moved to the repository's `Percolation_viability/` directory (`percolation_path.py`, `network_dimensionality.py`, `periodic_mincut.` and their tests `tests/test_percolation_path.py`, `tests/test_network_dimensionality.py`, `tests/test_periodic_mincut.py`).

## L.1 Original objective (mission #1)

The very first report (6 toy compounds) established that the percolation descriptor computed by `percolation_path.py` diverges sharply from classic ICOHP aggregates (sum, mean, min, max), but without a sample large enough to statistically test whether that difference is *useful*: does the percolation descriptor carry predictive information about thermodynamic stability (`energy_above_hull`) that the classic aggregates don't already carry? Campaign 2 built a 186-compound dataset (180 new + the 6 pilot compounds) split into three families (`exp_stable`, `exp_metastable`, `theo_metastable`, §??) to settle that question.

## L.2 Original workflow diagram



## L.3 Campaign 2 dataset (186 compounds)

| Family | $n$ | $E_{\text{hull}}$ range (eV/atom) |
|---|---|---|
| exp_stable | 63 | 0.0000 (by construction) |
| exp_metastable | 63 | 0.0005 – 0.0907 |
| theo_metastable | 60 | 0.0013 – 0.1991 |

| Bond type (`classify()`) | $n$ |
|---|---|
| metallic | 88 |
| covalent | 23 |
| ionic | 9 |
| unclassified | 66 |

35% of compounds (66/186) fall outside `classify()`'s three categories — mostly transition-metal pnictides/chalcogenides and hydrogen-containing compounds. (Historical snapshot of campaign 2 alone; the up-to-date table across the current 349 structures is in §**??**.)

## L.4 Results (with the percolation weight)

| Group | Metric | $n$ | $\rho$ | $p$ |
|---|---|---|---|---|
| all | percolation weight (raw) | 186 | 0.064 | 0.383 |
| all | percolation weight (normalized) | 186 | 0.071 | 0.339 |
| metallic | percolation weight (raw) | 88 | 0.191 | 0.075 |
| metallic | percolation weight (normalized) | 88 | **0.222** | **0.038** |
| ionic* | percolation weight (raw) | 9 | −0.402 | 0.284 |
| covalent | percolation weight (raw) | 23 | 0.165 | 0.453 |
| all | ICOHP sum | 186 | 0.032 | 0.670 |
| all | ICOHP min | 186 | 0.093 | 0.209 |
| metallic | ICOHP sum | 88 | 0.223 | 0.037 |
| metallic | ICOHP min | 88 | 0.198 | 0.065 |

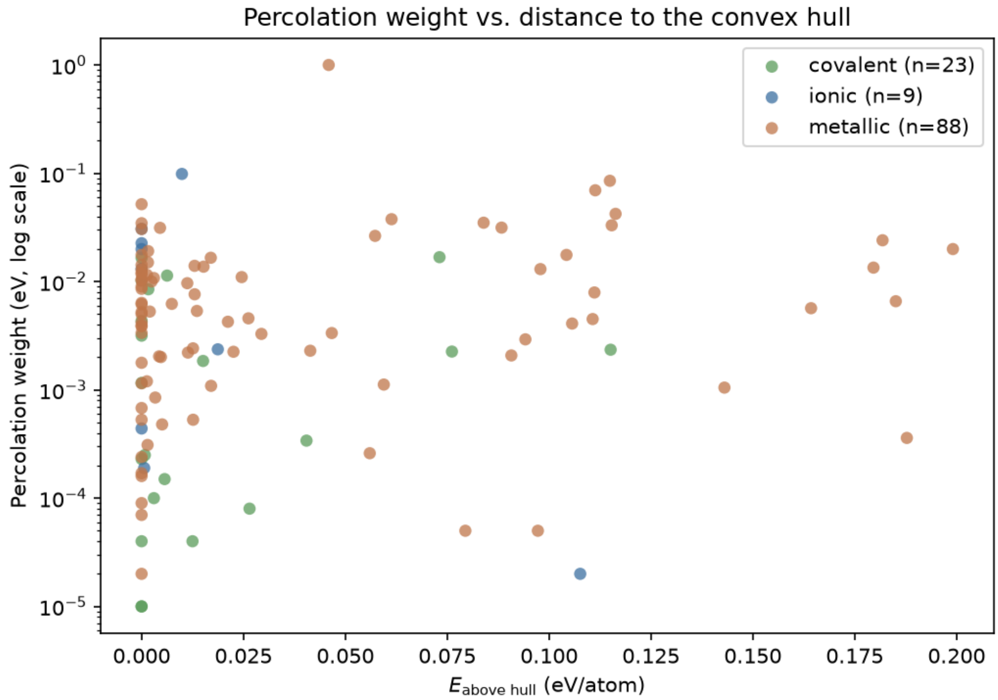
* $n < 15$: do not over-interpret.

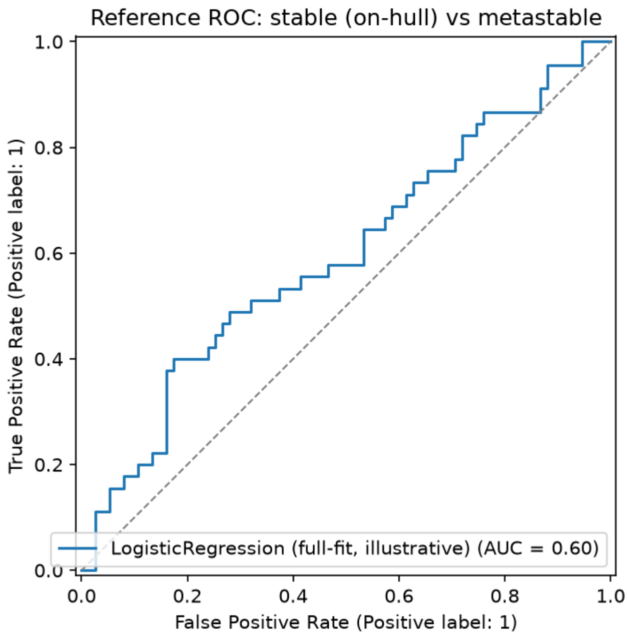
In the one subgroup where any result reaches nominal significance (metallic, $n = 88$), the normalized percolation weight ($\rho = 0.222$, $p = 0.038$) and the ICOHP sum ($\rho = 0.223$, $p = 0.037$) are statistically indistinguishable in strength. **The percolation descriptor does not demonstrate additional predictive power over the classic aggregates on this dataset**.

**Reference logistic regression.** Stable (on-hull) vs. metastable, features: normalized percolation weight, mean ICOHP, bond-type dummies ($n = 120$ after dropping unclassified compounds).

| | |
|---|---|
| AUC (5-fold cross-validation) | $0.496 \pm 0.141$ |
| per-fold AUC | 0.556, 0.430, 0.252, 0.637, 0.607 |

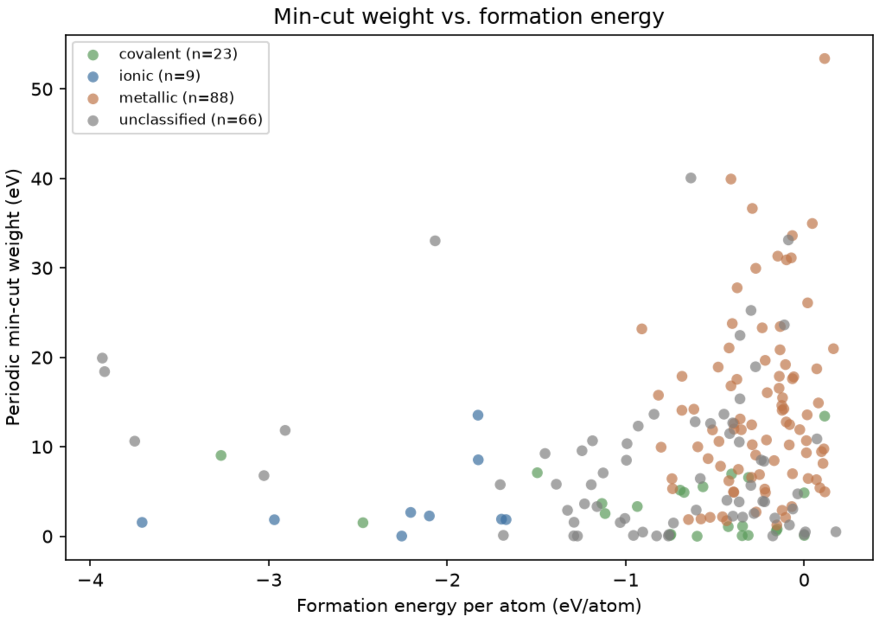
AUC $\approx 0.5$: chance level. The wide spread across folds (0.25–0.64) is itself informative: it indicates the absence of a stable, learnable signal at this sample size, not merely a "weak but real" one.




## L.5 Mission #3: network dimensionality and periodic min-cut

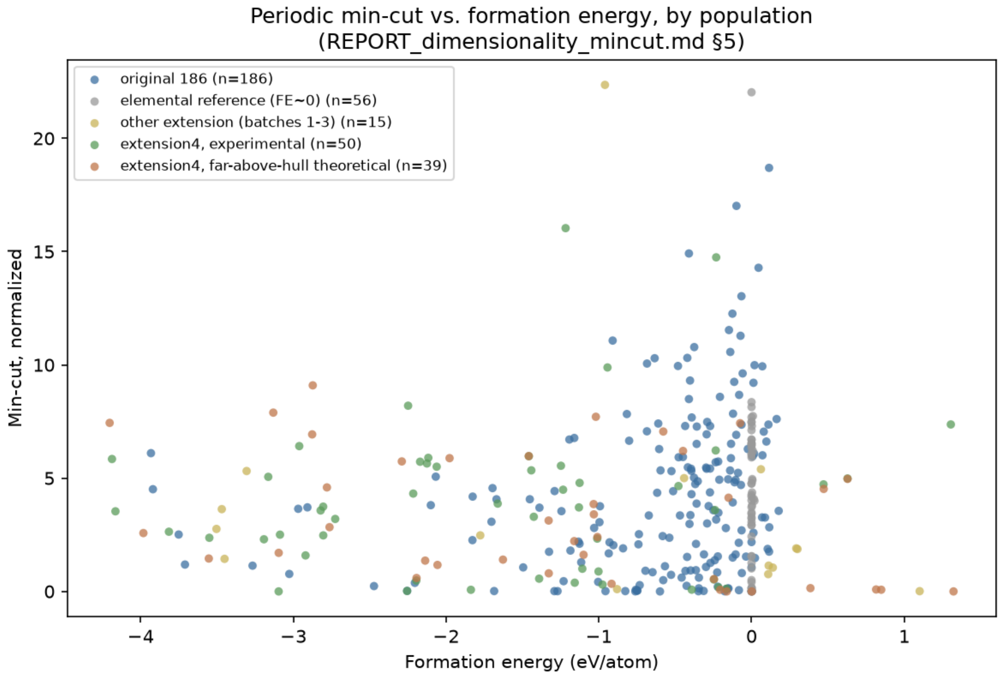
Two new descriptors, `percolation_path.py` left unchanged: `network_dimensionality.py` (0D–3D classification via a relative bond-strength threshold + breadth-first search) and `periodic_mincut.py` (minimum cut via `networkx` max-flow, separability rather than traversability). Headline: the normalized min-cut is the **first descriptor in the project to reach a significant global correlation** ($\rho = 0.285$, $p = 1.0 \times 10^{-4}$, $n = 186$), likely driven in part by between-bond-type clustering; dimensionality alone does not separate `formation_energy_per_atom` (Kruskal–Wallis $p = 0.19$).




**Why did the min-cut correlation collapse?** Mission #3's headline, $\rho = 0.285$ at $n = 186$, collapsed to $\rho = 0.089$, $p = 0.098$ (not significant) at $n = 349$. **This is not a refutation of the original result** — isolated, the original 186 compounds still give $\rho = 0.2845$, $p = 8.3 \times 10^{-5}$, essentially identical to the mission-#3 number. The collapse is a dilution artifact from pooling populations added for unrelated reasons:

- 65 of the 74 extension-batch-1–3 compounds are single-element reference structures (added for mission #5's elemental references), sitting at `formation_energy_per_atom`$\approx 0$ by construction with widely-scattered min-cut values — mechanically dilutes a rank correlation regardless of any true relationship. Excluding them recovers $\rho = 0.2157$ ($n = 193$).

- `extension4` is not one population but two, with *opposite-sign* correlations: the experimental half is flat ($\rho = 0.06$, not significant, $n = 50$); the deliberately far-above-hull theoretical half shows a *significant* correlation of the *opposite* sign ($\rho = -0.40$, $p = 0.011$, $n = 39$) — a higher min-cut there is associated with a *more* negative (more stable) formation energy, the reverse of the original population's trend.




**This opposite sign is itself resolved** (August-16 follow-up, four independent checks on the 39 `theo_far_hull` compounds, `analysis/investigate_theo_far_hull_signflip.py`): neither a cell-size/symmetry selection artifact (survives controlling for `n_sites`+`sg_number`, $\rho = -0.39$) nor a real physical effect of far-from-hull structures, but a **between-anion-chemistry confound** — F-containing systems (high min-cut, strongly negative formation energy) and azide $N_3$-containing systems (near-zero min-cut, near-zero/positive formation energy) pull the pooled correlation in opposite directions. Controlling for anion identity (dummy regression, $n = 39$) or comparing the experimental/theoretical pair within the exact same chemical system ($n = 37$ pairs) both independently collapse the correlation to $\rho \approx -0.07$, not significant — the same between-group clustering failure mode this project has now hit three times (min-cut/`bond_type`, antibonding/`is_metal`, and now min-cut/anion).

**Recommendation**: any future min-cut-vs-formation-energy number (or, by the same logic, any other descriptor's) should be conditioned on `family`/selection-provenance, the same way this project already conditions on `bond_type`/`is_metal`.

## L.6 Limits specific to the percolation weight

- **Primitive vs. conventional cell**: not reprocessed on the 186-compound dataset. Most likely explanation for the weak observed correlations — the strongest bond can sit entirely inside the primitive cell and contribute to no non-contractile cycle at all, biasing the measured "weakest link." Tested on the 6 pilot compounds (mission #2): the choice changes the percolation weight substantially (3x-55x) but in the *opposite* direction from the hypothesis (the weight gets *smaller* with a larger cell); not extended to the full dataset based on this result.

- **Chance-level reference logistic regression** (AUC $\approx 0.5$) with percolation weight as a feature.

- **Why not SISSO**: a single candidate descriptor (the percolation weight, in two algebraically related variants), compared against aggregates that are themselves functions of the same underlying ICOHP values — no meaningful combinatorial search space at this stage.

## L.7 Conclusion of the original campaign

The 186-compound dataset was built, computed (VASP+LOBSTER), and analyzed end to end. The headline result is a *negative* but quantified and honest one: at this sample size, the percolation descriptor shows no significant correlation with thermodynamic stability, and demonstrates no advantage over classic ICOHP aggregates. This result, together with the min-cut result (mission #3), motivated the project's reframing around Δ(ICOHP) documented in §**??**.

## L.8 Compound List (percolation weight)

Campaign 2's 186 compounds, grouped by category (family), with their Materials Project identifier, PBE thermodynamic hull distance (`energy_above_hull`), and the minimum ICOHP percolation weight (`icohp_percolation_weight_min`). Auto-generated from `analysis/percolation_vs_hull.csv` by `report/gen_appendix.py`.

Table S18: Complete list of the 186 compounds, grouped by category, with their Materials Project identifier, PBE hull distance, and minimum ICOHP percolation weight.

| Formula | mp-id | Bonding | $E_{\mathrm{hull}}$ (eV/at) | Percolation weight (eV) |
|---|---|---|---|---|
| **Experimental, stable (hull)** ($n = 63$) | | | | |
| As2S3 | mp-1189572 | covalent | 0.0000 | 1.00e-05 |
| AsF3 | mp-28027 | covalent | 0.0000 | 0.00e+00 |
| BI3 | mp-23189 | covalent | 0.0000 | 1.00e-05 |
| BW | mp-7832 | – | 0.0000 | 1.90e-04 |
| BaAu2 | mp-30363 | metallic | 0.0000 | 5.19e-02 |
| BaCl2 | mp-568662 | ionic | 0.0000 | 1.31e-02 |
| BaH2 | mp-23715 | – | 0.0000 | 2.40e-04 |
| Be2Cu | mp-2031 | metallic | 0.0000 | 1.15e-03 |
| Be2Nb3 | mp-11272 | metallic | 0.0000 | 6.36e-03 |
| Be2V | mp-11281 | metallic | 0.0000 | 6.80e-04 |
| CaAl4 | mp-1749 | metallic | 0.0000 | 1.03e-02 |
| CaSn | mp-2450 | metallic | 0.0000 | 5.23e-03 |
| CdPd | mp-1696 | metallic | 0.0000 | 9.00e-05 |
| CoB | mp-20857 | – | 0.0000 | 6.00e-05 |
| Cs2As3 | mp-15557 | – | 0.0000 | 2.02e-02 |
| FeS | mp-505531 | – | 0.0000 | 3.05e-03 |
| Ge2Os | mp-10032 | metallic | 0.0000 | 1.21e-02 |
| HfTc2 | mp-1095669 | metallic | 0.0000 | 1.77e-02 |
| Hg4Pt | mp-936 | metallic | 0.0000 | 3.36e-03 |
| HgF2 | mp-8177 | – | 0.0000 | 8.95e-03 |
| InBr | mp-22870 | covalent | 0.0000 | 1.02e-02 |
| KI | mp-22898 | ionic | 0.0000 | 2.26e-02 |
| KN3 | mp-827 | – | 0.0000 | 2.80e-04 |
| Li2Se | mp-2286 | – | 0.0000 | 1.16e-02 |
| LiB | mp-1001835 | – | 0.0000 | 2.91e-03 |
| Mg2Ni | mp-2137 | metallic | 0.0000 | 1.60e-04 |
| MgP4 | mp-384 | – | 0.0000 | 0.00e+00 |
| MnAl12 | mp-1104165 | metallic | 0.0000 | 1.42e-02 |
| MoSe2 | mp-1634 | covalent | 0.0000 | 1.16e-03 |
| NaS | mp-2400 | – | 0.0000 | 0.00e+00 |
| Na | mp-10172 | – | 0.0000 | 1.97e-03 |
| Nb3Ir | mp-1458 | metallic | 0.0000 | 9.02e-03 |
| NbGa3 | mp-1973 | metallic | 0.0000 | 8.58e-03 |
| PI2 | mp-29443 | covalent | 0.0000 | 2.30e-04 |
| PbSe | mp-2201 | covalent | 0.0000 | 1.66e-02 |

Table S18 (continued)

| Formula | mp-id | Bonding | $E_{\mathrm{hull}}$ (eV/at) | Percolation weight (eV) |
|---|---|---|---|---|
| RbI | mp-22903 | ionic | 0.0000 | 2.00e-02 |
| Sc5Ge3 | mp-17190 | metallic | 0.0000 | 5.30e-04 |
| ScAg2 | mp-1018128 | metallic | 0.0000 | 1.70e-04 |
| ScTe | mp-10026 | – | 0.0000 | 7.12e-03 |
| SnPt | mp-19856 | metallic | 0.0000 | 1.29e-02 |
| SrO2 | mp-2697 | ionic | 0.0000 | 4.40e-04 |
| SrSi | mp-2661 | metallic | 0.0000 | 7.00e-05 |
| Ta5Ge3 | mp-17593 | metallic | 0.0000 | 3.46e-02 |
| TaP | mp-1067587 | – | 0.0000 | 3.14e-02 |
| TaSb2 | mp-11697 | metallic | 0.0000 | 1.17e-02 |
| Te2Pd | mp-782 | – | 0.0000 | 1.44e-02 |
| Te2Ru | mp-267 | covalent | 0.0000 | 4.34e-03 |
| TeO2 | mp-8377 | covalent | 0.0000 | 4.00e-05 |
| Ti3In | mp-866046 | metallic | 0.0000 | 5.01e-03 |
| Ti3Pt | mp-2339 | metallic | 0.0000 | 1.78e-03 |
| TiBe12 | mp-1104067 | metallic | 0.0000 | 2.40e-04 |
| TiNi | mp-1048 | metallic | 0.0000 | 3.91e-03 |
| TiO | mp-1071163 | – | 0.0000 | 6.40e-04 |
| TiSi2 | mp-1077503 | metallic | 0.0000 | 6.18e-03 |
| TlCl | mp-571079 | – | 0.0000 | 7.17e-03 |
| VNi2 | mp-11531 | metallic | 0.0000 | 3.88e-03 |
| WO2 | mp-1524394 | – | 0.0000 | 2.20e-04 |
| Zr2Au | mp-2348 | metallic | 0.0000 | 3.06e-02 |
| Zr5Pb3 | mp-681992 | metallic | 0.0000 | 2.00e-05 |
| ZrMo2 | mp-2049 | metallic | 0.0000 | 1.05e-02 |
| Si | mp-149 | covalent | 0.0000 | 3.17e-03 |
| NaCl | mp-22862 | ionic | 0.0000 | 3.07e-02 |
| AlNi | mp-1487 | metallic | 0.0000 | 4.19e-03 |
| **Experimental, metastable** ($n = 63$) | | | | |
| CdI2 | mp-567259 | – | 0.0005 | 2.38e-02 |
| MgCl2 | mp-570259 | ionic | 0.0006 | 1.90e-04 |
| C | mp-169 | covalent | 0.0008 | 2.50e-04 |
| ZrBr | mp-1065846 | – | 0.0008 | 7.88e-03 |
| Li3Hg | mp-1646 | metallic | 0.0013 | 1.20e-03 |
| Ta7Co6 | mp-1104548 | metallic | 0.0015 | 3.10e-04 |
| CaGa4 | mp-2461 | metallic | 0.0016 | 1.91e-02 |
| InCl | mp-571636 | covalent | 0.0016 | 8.50e-03 |
| NiB | mp-14019 | – | 0.0018 | 4.10e-04 |
| MgH2 | mp-23711 | – | 0.0019 | 1.00e-05 |
| Sn7Ir5 | mp-20466 | metallic | 0.0024 | 1.00e-02 |
| NaN3 | mp-1066400 | – | 0.0029 | 7.00e-04 |
| ClF3 | mp-556767 | – | 0.0030 | 0.00e+00 |
| SiS2 | mp-1095294 | covalent | 0.0030 | 1.00e-04 |
| Sr5Sn3 | mp-17720 | metallic | 0.0030 | 1.08e-02 |
| Te2Ir | mp-1551 | – | 0.0030 | 2.80e-03 |
| Li13Sn5 | mp-30769 | metallic | 0.0033 | 8.50e-04 |
| H3N | mp-643432 | covalent | 0.0038 | 0.00e+00 |
| Cs2Se | mp-1011696 | – | 0.0042 | 2.62e-02 |
| In3Ir | mp-630976 | metallic | 0.0043 | 2.04e-03 |
| HI | mp-697025 | – | 0.0044 | 0.00e+00 |
| Rb2Hg7 | mp-31474 | metallic | 0.0045 | 3.14e-02 |
| PbAu2 | mp-19871 | metallic | 0.0047 | 2.01e-03 |
| BeCu | mp-2323 | metallic | 0.0050 | 4.80e-04 |
| SiO2 | mp-546794 | covalent | 0.0056 | 1.50e-04 |

Table S18 (continued)

| Formula | mp-id | Bonding | $E_{\mathrm{hull}}$ (eV/at) | Percolation weight (eV) |
|---|---|---|---|---|
| IF7 | mp-27988 | – | 0.0063 | 0.00e+00 |
| K | mp-10157 | – | 0.0069 | 8.07e-02 |
| Al3Os2 | mp-16521 | metallic | 0.0074 | 6.23e-03 |
| Nb2O5 | mp-1595 | – | 0.0075 | 1.00e-04 |
| LiBr | mp-23259 | ionic | 0.0099 | 9.88e-02 |
| ZnAg | mp-1912 | metallic | 0.0112 | 9.67e-03 |
| MgPd3 | mp-7753 | metallic | 0.0114 | 2.21e-03 |
| AsPd2 | mp-1080831 | – | 0.0118 | 3.50e-04 |
| GeSe2 | mp-10074 | covalent | 0.0125 | 4.00e-05 |
| Re3Ge7 | mp-29141 | metallic | 0.0126 | 2.42e-03 |
| Cu3Ge | mp-19724 | metallic | 0.0126 | 5.30e-04 |
| Ti3Ge | mp-1189044 | metallic | 0.0130 | 1.40e-02 |
| ZrGa3 | mp-30685 | metallic | 0.0130 | 7.64e-03 |
| SrAl2 | mp-1638 | metallic | 0.0136 | 5.36e-03 |
| NiS | mp-1547 | – | 0.0142 | 3.00e-04 |
| Al12W | mp-11227 | metallic | 0.0152 | 1.37e-02 |
| CsH | mp-632319 | – | 0.0161 | 2.66e-02 |
| CaGa2 | mp-992 | metallic | 0.0170 | 1.66e-02 |
| NiHg4 | mp-570835 | metallic | 0.0171 | 1.09e-03 |
| SiRh | mp-2287 | metallic | 0.0246 | 1.10e-02 |
| BaSn5 | mp-571353 | metallic | 0.0262 | 4.58e-03 |
| PS | mp-612 | covalent | 0.0265 | 8.00e-05 |
| AsRh | mp-22079 | – | 0.0269 | 1.93e-03 |
| Te2Au | mp-567525 | – | 0.0274 | 1.62e-02 |
| Be3N2 | mp-6977 | – | 0.0315 | 2.00e-05 |
| Y2O3 | mp-558573 | – | 0.0342 | 5.00e-05 |
| AgO | mp-1079720 | – | 0.0423 | 4.00e-05 |
| FeTe2 | mp-1102002 | – | 0.0449 | 5.20e-04 |
| H2O | mp-2739191 | – | 0.0494 | 0.00e+00 |
| CdO2 | mp-2310 | – | 0.0562 | 9.00e-05 |
| Mo3Se4 | mp-21021 | – | 0.0609 | 1.84e-03 |
| PbSe | mp-22009 | covalent | 0.0731 | 1.68e-02 |
| Be12Pd | mp-12666 | metallic | 0.0795 | 5.00e-05 |
| Sr2As | mp-15697 | – | 0.0800 | 2.06e-03 |
| AsRh2 | mp-1102364 | – | 0.0806 | 4.63e-03 |
| ZrRh | mp-2808 | metallic | 0.0839 | 3.50e-02 |
| K2S | mp-559278 | – | 0.0873 | 1.07e-02 |
| Ge4Rh | mp-1104286 | metallic | 0.0907 | 2.08e-03 |
| **Theoretical, metastable** ($n = 60$) | | | | |
| Li3Ag | mp-976408 | metallic | 0.0013 | 1.15e-02 |
| Y3Co | mp-1189329 | metallic | 0.0016 | 1.50e-02 |
| TiNi | mp-1067248 | metallic | 0.0020 | 5.29e-03 |
| TaP | mp-1187244 | – | 0.0033 | 3.59e-02 |
| Sr3As4 | mp-678007 | – | 0.0062 | 4.47e-03 |
| Ga2Se3 | mp-1224809 | covalent | 0.0063 | 1.14e-02 |
| MnN | mp-1009130 | covalent | 0.0151 | 1.85e-03 |
| BeCl2 | mp-1190080 | ionic | 0.0187 | 2.37e-03 |
| MgSn | mp-1094801 | metallic | 0.0212 | 4.26e-03 |
| TiW | mp-1216621 | metallic | 0.0225 | 2.25e-03 |
| SrTl3 | mp-1206636 | metallic | 0.0294 | 3.29e-03 |
| IrCl3 | mp-729507 | – | 0.0301 | 2.72e-03 |
| Ta4C3 | mp-1218000 | – | 0.0335 | 1.44e-03 |
| Sc2O3 | mp-755313 | – | 0.0364 | 5.10e-04 |
| SnBr2 | mp-978985 | covalent | 0.0405 | 3.40e-04 |

Table S18 (continued)

| Formula | mp-id | Bonding | $E_{hull}$ (eV/at) | Percolation weight (eV) |
|---|---|---|---|---|
| Sn7Pd5 | mp-1219006 | metallic | 0.0414 | 2.30e-03 |
| Tl4Sn | mp-1216560 | metallic | 0.0459 | 1.00e+00 |
| LiTl3 | mp-1185253 | metallic | 0.0467 | 3.35e-03 |
| PtCl2 | mp-1219748 | – | 0.0490 | 5.04e-03 |
| HCl | mp-684609 | – | 0.0536 | 2.50e-04 |
| MgSn | mp-1094669 | metallic | 0.0560 | 2.60e-04 |
| K5As4 | mp-684799 | – | 0.0564 | 8.80e-03 |
| KSn3 | mp-1185151 | metallic | 0.0573 | 2.65e-02 |
| NbS2 | mp-774873 | – | 0.0576 | 1.20e-04 |
| YAg3 | mp-1187811 | metallic | 0.0594 | 1.12e-03 |
| B3Ir2 | mp-1228647 | – | 0.0609 | 2.80e-04 |
| TiMo2 | mp-1216675 | metallic | 0.0613 | 3.77e-02 |
| Cr2C | mp-1226378 | – | 0.0625 | 2.08e-03 |
| ZrO2 | mp-775909 | – | 0.0627 | 1.20e-04 |
| Zn | mp-2647117 | – | 0.0717 | 1.04e-02 |
| IN2 | mp-1097011 | – | 0.0730 | 4.70e-04 |
| CoSe2 | mp-1390196 | – | 0.0734 | 8.40e-04 |
| GeS | mp-12910 | covalent | 0.0762 | 2.26e-03 |
| ZrTi | mp-1215236 | metallic | 0.0883 | 3.15e-02 |
| VCo3 | mp-1095057 | metallic | 0.0942 | 2.93e-03 |
| Tl3Ga | mp-1187539 | metallic | 0.0972 | 5.00e-05 |
| CaGe2 | mp-643542 | metallic | 0.0979 | 1.30e-02 |
| NbRh2 | mp-1220414 | metallic | 0.1043 | 1.76e-02 |
| TcW | mp-1217337 | metallic | 0.1056 | 4.10e-03 |
| MgF2 | mp-560236 | ionic | 0.1076 | 2.00e-05 |
| VN | mp-1017532 | – | 0.1080 | 4.00e-05 |
| Na2Cl | mp-990084 | – | 0.1105 | 3.76e-02 |
| ZnPb3 | mp-1187949 | metallic | 0.1107 | 4.51e-03 |
| Sr3Sn | mp-1187136 | metallic | 0.1111 | 7.94e-03 |
| TaRe | mp-1217894 | metallic | 0.1113 | 6.98e-02 |
| Al4Si | mp-1228120 | metallic | 0.1149 | 8.54e-02 |
| B6Pb | mp-1096825 | covalent | 0.1151 | 2.35e-03 |
| ScTc3 | mp-867262 | metallic | 0.1154 | 3.31e-02 |
| CrMo | mp-1226200 | metallic | 0.1163 | 4.23e-02 |
| BPd5 | mp-1228665 | – | 0.1325 | 2.42e-02 |
| Zr3Be | mp-1188016 | metallic | 0.1430 | 1.05e-03 |
| Na3Cl | mp-1064484 | – | 0.1456 | 2.55e-03 |
| SiAu3 | mp-1186998 | metallic | 0.1643 | 5.68e-03 |
| InSe2 | mp-1232036 | – | 0.1733 | 1.64e-03 |
| TiSi2 | mp-570991 | metallic | 0.1796 | 1.35e-02 |
| BeAu3 | mp-983539 | metallic | 0.1818 | 2.41e-02 |
| Zr3Ge | mp-1188039 | metallic | 0.1851 | 6.59e-03 |
| AlSb | mp-1023918 | metallic | 0.1878 | 3.60e-04 |
| Mg15B | mp-1023515 | – | 0.1943 | 2.80e-04 |
| HfMo | mp-1224314 | metallic | 0.1991 | 2.00e-02 |